Markovian Anticipated BSDEs, Their Feynman-Kac Representation and the Convergence Rate of Their Time-Discretization

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Abstract

In this work, we introduce and study a class of anticipated backward stochastic differential equations (ABSDEs) whose driver depends on a measure-valued random function of the solution. This framework unifies and extends previous works on ABSDEs, including those where the dependence occurs through conditional expectations or conditional risk measures. Under Lipschitz and square integrability assumptions in a Brownian filtration, we establish a priori estimates, existence and uniqueness of square integrable solutions. In the Markovian case, we prove that the anticipated term inherits the Markov structure and that the solution admits a Feynman-Kac representation in terms of a deterministic function of the underlying factor process.We formulate an associated PDE, nonlocal in time and space, and prove a verification theorem showing that any sufficiently smooth solution of this PDE coincides with the ABSDE value function. Finally, we show that an implicit ABSDE time-discretization scheme achieves the classical square-root convergence rate.

Keywords: Anticipated BSDEs, Conditional distribution, Feynman-Kac formula, Nonlocal PDEs, Derivative pricing, Risk Management.

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1 Introduction

Classical BSDEs

$$Y_t = \xi + \int_t^T f(s, Y_s, Z_s)ds - \int_t^T Z_s dW_s, \quad 0 \le t \le T, \tag{1}$$

where ξ is the terminal condition, f is the driver, and (Y, Z) the solution, are widely studied since the pioneer work of Pardoux and Peng (1990), see also El Karoui et al. (1997). Here, the driver only depends on the values of the solution at current time, making the equation "local in time". A key result is the Feynman-Kac representation: when the data ξ and f are Markovian, i.e. functions of an underlying Markov process X, the solution can be represented as $Y_t = u(t, X_t)$ where the value function u solves a semilinear parabolic partial differential equation.

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ABSDEs arise in several applications. Peng and Yang (2009) initially introduced them for dealing with stochastic optimal control problems with delay. Further papers in this direction include Chen and Wu (2010) and Meng et al. (2025). On the financial side, regulated financial institutions must hold collateral and capital against their potential future losses. Basel Committee on Banking Supervision (2013) requires banks to compute their collateral and capital based on value-at-risk or expected shortfall of these future losses. On top of the traditional credit valuation adjustment (CVA) for counterparty default risk, the pricing of derivative instruments incorporates various valuation adjustments (XVAs) including margin valuation adjustment (MVA) for the cost of having to raise segregated collateral in the form of initial margin, funding valuation adjustment (FVA) for the cost of having to raise rehypothecable collateral in the form of variation margin, and capital valuation adjustment (KVA) for the cost of having to remunerate shareholders for their capital at risk. When accounted for in a dynamic XVA setup, collateral and capital loop lead to XVA ABSDEs. The MVA equation of Agarwal et al. (2019) is thus already an ABSDE. As reviewed in Crépey (2026, Chapter VII) building on Crépey et al. (2020), the fact that the KVA reserve is eligible to capital at risk, itself usable as variation margin, leads to a system of ABSDEs for the FVA and the KVA.

With these motivations in mind, we consider in this paper a class of ABSDEs, encompassing both Peng and Yang (2009) and Crépey et al. (2020), whose driver at time t depends on a measure-valued random function of the restriction to $[t, T+\theta]$ of the processes Y and Z. By comparison, Agarwal et al. (2019) anticipate through a real valued random function of the restriction to $[t, T+\theta]$ of the process Y, Yang and Elliott (2013) through a functional dependence on future realized trajectories of (Y, Z) ('ω by ω'), Cheridito and Nam (2017) and Klimsiak and Rzymowski (2023) through abstract fixed-point or martingale-part formulations, and Hu, Li, and Wen (2026) through conditional expectations in a multi-dimensional quadratic setting.

Under Lipschitz conditions, in a semimartingale setup driven by Brownian motions, a careful analysis of the conditional distributions and fixed-point arguments yield a priori estimates, existence and uniqueness of the solution. In the Markovian case, we establish a Feynman-Kac representation in terms of a deterministic function of the underlying factor process. We then introduce a PDE, nonlocal not only in space (as for classical BSDEs in jump-diffusion models or with McKean-Vlasov BSDEs) but also in time, and we prove by a verification argument that any sufficiently smooth solution of this PDE coincides with the ABSDE value function. We last prove the convergence, at a square-root rate of an implicit time-discretization ABSDE scheme.

The rest of the paper is organized as follows. Section 2 develops our fundamental ABSDE results including Proposition 2.4 for a priori estimates, Theorem 2.7 on existence and uniqueness of solutions. Section 3 specializes these results to the Markovian case, Theorem 3.6 showing that $Y_t = u(t, X_t)$ then holds for some deterministic function u, Hölder-regular in time and Lipschitz-continuous in space, and Proposition 3.7 establishes the connection to the nonlocal PDE (30): it verifies that a smooth solution (if any) to the PDE coincides with the value function u. Section 4 proves that the implicit time-discretization scheme for classical BSDEs extends to the anticipated framework, without any loss in convergence rate. Section 5 gives explicit calculations of the scheme, and Section 6 presents numerical results for the MVA. Section 7 concludes the paper with a discussion of open questions and future research directions.

The main contributions with respect to previous literature are thus: on the stochastic analysis side, a Feynman-Kac representation for Markovian ABSDEs, in terms of a value function u defined on the state-space of the factor process itself (no space-lifting required despite the dependence of the coefficient on the conditional distribution of the solution in the future), linked by a verification argument to a PDE nonlocal in time and space; on the numerical analysis side, a proof of convergence for the implicit Markovian ABSDEs time-discretization scheme of Abbas-Turki et al. (2025), at the square-root rate of classical BSDEs.

1.1 Notations

We denote by $|\cdot|$ the Euclidean norm of a vector (or matrix) in the dimension(s) of its argument. Given nonnegative constants T (a maturity) and θ (a terminal delay) and a filtered probability space $(\Omega, \mathcal{A}, \mathfrak{F} = (\mathfrak{F}_t)_{0\le t\le T+\theta}, \mathbb{P})$, where the filtration $\mathfrak{F}$ satisfies the usual conditions, we denote by $Prog$ the progressive sigma algebra on $\Omega \times [0,T]$ and we consider the following spaces of m-dimensional processes U, given positive integers $m, p \ge 1$:

$$\mathcal{H}^p(\mathfrak{F}, \mathbb{R}^m) = \left\{ \mathfrak{F} \text{ progressive processes } U \,;\, \|U\|^p_{\mathcal{H}^p} = \mathbb{E}\left[\int_0^{T+\theta} |U_t|^p dt\right] < \infty \right\},$$
$$\mathcal{S}^2(\mathfrak{F}, \mathbb{R}^m) = \left\{ \mathfrak{F} \text{ adapted continuous processes } U \,;\, \|U\|^2_{\mathcal{S}^2} = \mathbb{E}\left[\sup_{0\le t\le T+\theta} |U_t|^2\right] < \infty \right\}.$$

As usual, processes in $\mathcal{H}^p$ are identified up to $dt \otimes d\mathbb{P}$-a.e. equality, whereas processes in $\mathcal{S}^2$ are identified up to indistinguishability, so that $\|\cdot\|_{\mathcal{H}^2}$ and $\|\cdot\|_{\mathcal{S}^2}$ define respective norms on $\mathcal{H}^2(\mathfrak{F}, \mathbb{R}^m)$ and $\mathcal{S}^2(\mathfrak{F}, \mathbb{R}^m)$ and that, for any $\beta \ge 0$,

$$\|U\|_{\mathcal{H}^2_\beta} = \mathbb{E}\left[\int_0^{T+\theta} |U_t|^2 e^{\beta t} dt\right]^{\frac{1}{2}} \quad \text{and } \|U\|_{\mathcal{S}^2_\beta} = \mathbb{E}\left[\sup_{0\le t\le T+\theta} |U_t|^2 e^{\beta t}\right]^{\frac{1}{2}}$$

define norms respectively equivalent to $\|\cdot\|_{\mathcal{H}^2} \equiv \|\cdot\|_{\mathcal{H}^2_0}$ on $\mathcal{H}^2(\mathfrak{F}, \mathbb{R}^m)$ and to $\|\cdot\|_{\mathcal{S}^2} \equiv \|\cdot\|_{\mathcal{S}^2_0}$ on $\mathcal{S}^2(\mathfrak{F}, \mathbb{R}^m)$. For $\beta \ge 0$, $0 \le s \le t \le T+\theta$ and for $U \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^m)$, resp. $\mathcal{S}^2(\mathfrak{F}, \mathbb{R}^m)$, let also

$$\|U\|_{\mathcal{H}^2_\beta(s,t)} = \mathbb{E}\left[\int_s^t |U_r|^2 e^{\beta r} dr\right]^{\frac{1}{2}} \text{ and } \|U\|_{\mathcal{H}^2_\beta(t)} = \|U\|_{\mathcal{H}^2_\beta(0,t)} \text{ , resp.}$$
$$\|U\|_{\mathcal{S}^2_\beta(s,t)} = \mathbb{E}\left[\sup_{s\le r\le t} |U_r|^2 e^{\beta r}\right]^{\frac{1}{2}} \text{ and } \|U\|_{\mathcal{S}^2_\beta(t)} = \|U\|_{\mathcal{S}^2_\beta(0,t)}.$$

We denote by $T_\sharp \mu$ the pushforward measure $\mu \circ (T^{-1})$ of a measure μ by a map T. Given a Polish space (E, d_E), we denote by $\mathcal{B}(E)$ the Borel sigma-algebra on E and by $\mathcal{P}(E)$ the set of probability measures on $(E, \mathcal{B}(E))$. For $p \ge 1$, we denote by $\mathcal{P}_p(E)$ the subset of probability measures $\mu \in \mathcal{P}(E)$ with finite pth moment, i.e. such that $\int_E d_E(e, e_0)^p \mu(de) < \infty$ holds for any fixed $e_0 \in E$. We endow $\mathcal{P}_p(E)$ with the p-Wasserstein distance defined by

$$W_p(\mu, \mu') = \left(\inf_{\pi \in \Pi(\mu,\mu')} \int_{E\times E} d_E(e, e')^p \pi(de, de')\right)^{\frac{1}{p}}, \ \mu, \mu' \in \mathcal{P}_p(E),$$

where $\Pi(\mu, \mu')$ is the set of probability measures on $E \times E$ with marginals μ and μ'. By Villani (2009, Theorem 6.18), $(\mathcal{P}_p(E), W_p)$ is a Polish space.

We write a.s. and a.e. for $d\mathbb{P}$ almost surely and $dt \otimes d\mathbb{P}$ almost everywhere (on $[0, T+\theta] \times \Omega$, unless otherwise stated). For a random variable ε defined on $(\Omega, \mathcal{A}, \mathbb{P})$ and taking values in a Polish space E, we denote by $\mathcal{L}(\varepsilon)$ its law, i.e. the probability measure on $(E, \mathcal{B}(E))$ defined by $\mathcal{L}(\varepsilon)(B) = \mathbb{P}(\varepsilon \in B)$, $B \in \mathcal{B}(E)$, and for any $p \ge 1$, we say that $\varepsilon \in L^p(\Omega; E)$ if $\mathcal{L}(\varepsilon) \in \mathcal{P}_p(E)$, meaning that for any $e_0 \in E$, we have $\mathbb{E}\left[d_E(\varepsilon, e_0)^p\right] < \infty$. For a sub-sigma-algebra $\mathcal{F}$ of $\mathcal{A}$, we denote by $\mathcal{L}(\varepsilon|\mathcal{F}) = \mathbb{P}(\varepsilon \in de|\mathcal{F})$ the conditional law of ε given $\mathcal{F}$, i.e. the probability measure on $\mathcal{B}(E)$ defined by $\mathcal{L}(\varepsilon|\mathcal{F})(B) = \mathbb{P}(\varepsilon \in B|\mathcal{F})$, $B \in \mathcal{B}(E)$. For any $t \le T+\theta$, we denote by $\mathbb{P}_t(A)$ the $(\mathfrak{F}_t, \mathbb{P})$ conditional probability of an event $A \in \mathcal{A}$, by $\mathbb{E}_t[\xi]$ the $(\mathfrak{F}_t, \mathbb{P})$ conditional expectation of an integrable real random variable ξ, and by $\mathcal{L}_t(\xi)$ the $(\mathfrak{F}_t, \mathbb{P})$ conditional law of an E valued $\mathcal{A}$ measurable random variable ξ, where E is Polish.

We denote by C (or $C(\rho)$) a generic positive constant (modulo perhaps dependence on parameters ρ but only on ρ) that may change from line to line.

2 BSDEs Anticipated in Conditional Distribution

After defining the anticipated BSDEs (ABSDEs) of interest in this work, we establish foundational results: a priori estimates, existence and uniqueness.

Hereafter $\mathfrak{F}$ is assumed to be the natural augmented filtration of a q variate Brownian motion $W = (W_t)_{0\leq t\leq T+\theta}$, for some positive integer q. We also denote by $\mathfrak{F}^T = (\mathfrak{F}_t)_{t\in[0,T]}$ the restriction of $\mathfrak{F}$ to $[0,T]$, for some maturity $T > 0$ and terminal delay $\theta \geq 0$.

Given a positive integer n , we consider the n variate ABSDE

$$\begin{aligned} &Z_t = \Psi_t,\ t \in (T, T+\theta] \text{ and, for } t \in [0, T+\theta], \\ &Y_t = \Phi_{t\vee T} + \int_{t\wedge T}^{T} f(s, Y_s, Z_s, m_s(Y,Z))ds - \int_{t\wedge T}^{T} Z_s dW_s, \end{aligned} \tag{2}$$

where

- (Φ, Ψ) is an $\mathbb{R}^n \times \mathbb{R}^{n\times q}$ valued $\mathfrak{F}$ adapted process,
- $f : \Omega \times [0,T] \times \mathbb{R}^n \times \mathbb{R}^{n\times q} \times \mathcal{P}_2(E) \to \mathbb{R}^n$ is a $Prog \otimes \mathcal{B}^n \otimes \mathcal{B}^{n\times q} \otimes \mathcal{B}(\mathcal{P}_2(E))$ measurable map, where E is a Polish space,
- $m : \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q}) \to \mathcal{H}^0(\mathfrak{F}, \mathcal{P}_2(E)), (Y,Z) \mapsto m(Y,Z) = (m_t(Y,Z))_{t\in[0,T]}$, with $\mathcal{H}^0(\mathfrak{F}, \mathcal{P}_2(E))$ the set of $\mathcal{P}_2(E)$ valued progressive processes, represents the anticipated term in the sense that, given $(Y,Z) \in \mathcal{H}^2(\mathfrak{F}; \mathbb{R}^n \times \mathbb{R}^{n\times q})$ and $t \in [0,T]$, the $\mathcal{P}_2(E)$ valued variable $m_t(Y,Z)(\omega)$ depends on the processes Y, Z at future times $s \in [t, T+\theta]$, but only via their $\mathfrak{F}_t$ conditional distribution (see Assumption 2.1(iii) and Remark 2.3 below).

Remark 2.1. Even if the terminal delay θ is equal to 0, at any time $t < T$, the anticipated term $m_t(Y,Z)$ still depends on the future (from t to T) of the solution, so the equation is still an anticipated BSDE. However, as t approaches T, the influence of the future path diminishes as the time-to-maturity $T - t$ decreases to 0. If $\theta > 0$, the influence of the future path remains significant up to the maturity.

We call *admissible kernel* ρ, any Borel kernel $[0,T]\times\mathcal{B}([0,T+\theta]) \ni (t,B) \mapsto \rho_t(B) \in \mathbb{R}_+$ with bounded mass $\sup_{t\in[0,T]} \rho_t([t,T+\theta]) \leq \Lambda_\rho$ for some $\Lambda_\rho \geq 0$, and such that, for every nonnegative Borel function $\varphi : [0, T+\theta] \to \mathbb{R}$,

$$\int_0^T \int_t^{T+\theta} \varphi_s \rho_t(ds)dt \leq \Lambda_\rho \int_0^{T+\theta} \varphi_s ds. \tag{3}$$

Both requirements are independent but needed in the sequel.

Example 2.2. Any kernel satisfying $\rho_t(ds) \leq Cds$ is admissible, as $\sup_{t\in[0,T]} \rho_t([t,T+\theta]) \leq \sup_{t\in[0,T]} C(T+\theta-t) = C(T+\theta)$, and for any nonnegative Borel function $\varphi : [0, T+\theta] \to \mathbb{R}$,

$$\int_0^T \int_t^{T+\theta} \varphi_s \rho_t(ds)dt \leq C\int_0^T \int_t^{T+\theta} \varphi_s ds dt \leq C\int_0^{T+\theta} \left(\varphi_s \int_0^{s\wedge T} dt\right) ds \leq CT\int_0^{T+\theta} \varphi_s ds.$$

The kernel $\rho_t(ds) = \delta_{t+\theta}(ds)$ is also admissible, as $\sup_{t\in[0,T]} \delta_{t+\theta}([t,T+\theta]) = 1$, and for any nonnegative Borel function $\varphi : [0, T+\theta] \to \mathbb{R}$,

$$\int_0^T \int_t^{T+\theta} \varphi_s \rho_t(ds)dt = \int_0^T \varphi_{t+\theta} dt = \int_\theta^{T+\theta} \varphi_s ds \leq \int_0^{T+\theta} \varphi_s ds.$$

We will say that (f, Φ, Ψ, m) are *ABSDE standard parameters* if they satisfy the following:

Assumption 2.1. **(i)** $(\Phi_t)_{T\le t\le T+\theta}$ is continuous, $(\Psi_t)_{T\le t\le T+\theta}$ is progressive, and

$$\mathbb{E}\left[\sup_{T\le s\le T+\theta}|\Phi_s|^2+\int_T^{T+\theta}|\Psi_s|^2ds\right]<\infty.$$

(ii) There exists a real constant Λ_f such that for any $(t,y,y',z,z',\mu,\mu')\in[0,T]\times(\mathbb{R}^n)^2\times(\mathbb{R}^{n\times q})^2\times\mathcal{P}_2(E)^2$,

$$\left|f(t,y,z,\mu)-f(t,y',z',\mu')\right|\le\Lambda_f\big(\left|y-y'\right|+\left|z-z'\right|+W_2(\mu,\mu')\big),\ \mathbb{P}\text{-a.s.},$$

where, as usual, the dependence of f on ω is suppressed from the notation. In addition, $f(\cdot,0,0,m_\cdot(0,0))\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n)$.
(iii) For any $(Y,Z),(Y',Z')\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})$, we have

$$W_2(m_t(Y,Z)(\omega),m_t(Y',Z'))^2\le\mathbb{E}_t\left[\int_t^{T+\theta}\big(\left|Y_s-Y'_s\right|^2+\left|Z_s-Z'_s\right|^2\big)\rho_t(ds)\right],\ \text{a.e.},\tag{4}$$

for some admissible kernel ρ.

Under Assumption 2.1 and in order to alleviate the notations, we denote $\Lambda:=1\vee\Lambda_f\vee\Lambda_\rho$.

The assumptions on Φ,Ψ and f are quite standard in BSDE setups.

Remark 2.3. If (Y,Z) and (Y',Z') coincide a.e. on $[t,T+\theta]$, for some $t\in[0,T]$, then, under Assumption 2.1(iii), $m_s(Y,Z)=m_s(Y',Z')$ a.s. for ds-almost every $s\in[t,T]$. Thus the values of (Y,Z) before s are irrelevant to compute $m_s(Y,Z)$, motivating the anticipated terminology.

Assumption 2.1(iii) is satisfied for a wide class of anticipated terms, as the following lemma (see also the examples below) shows.

Lemma 2.1. *Let there be given an admissible kernel ρ and a map $A:\mathcal{H}^2(\mathfrak{F};\mathbb{R}^n\times\mathbb{R}^{n\times q})\to L^2(\Omega\times[0,T],\mathfrak{F}_{T+\theta}\otimes\mathcal{B}([0,T]),\mathbb{P}\otimes dt;E)$ such that, for any $(Y,Z),(Y',Z')\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})^2$,*

$$\mathbb{E}_t\left[d_E(A(t,Y,Z),A(t,Y',Z'))^2\right]\le\mathbb{E}_t\left[\int_t^{T+\theta}\big(\left|Y_s-Y'_s\right|^2+\left|Z_s-Z'_s\right|^2\big)\rho_t(ds)\right]\tag{5}$$

holds a.s., for dt-almost every $t\in[0,T]$. Then there exists $m:\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})\to\mathcal{H}^0(\mathfrak{F},\mathcal{P}_2(E))$ such that Assumption 2.1(iii) is satisfied and, for any $(Y,Z)\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})$,

$$m_t(Y,Z)=\mathcal{L}(A(t,Y,Z)\mid\mathfrak{F}_t)\quad dt\otimes d\mathbb{P}\ a.e.\tag{6}$$

Moreover, under Assumption 2.1(ii), $\int_0^T|f(s,Y_s,Z_s,m_s(Y,Z))|ds<+\infty$ a.s.

Proof. We fix $(Y,Z)\in\mathcal{H}^2(\mathfrak{F};\mathbb{R}^n\times\mathbb{R}^{n\times q})$.

First, since $A(Y,Z)\in L^2(\mathbb{P}\otimes dt;E)$, we have for any $e_0\in E$,

$$\int_0^T\mathbb{E}[d_E(A(t,Y,Z),e_0)^2]dt<\infty.$$

In particular, for almost all $t\in[0,T]$, $\mathbb{E}\left[d_E(A(t,Y,Z),e_0)^2\right]<\infty$. For any such t, since

$$\mathbb{E}\left[\mathbb{E}\left[d_E(A(t,Y,Z),e_0)^2\mid\mathfrak{F}_t\right]\right]=\mathbb{E}\left[d_E(A(t,Y,Z),e_0)^2\right]<\infty,$$

we have $\mathbb{E}\left[d_E(A(t,Y,Z),e_0)^2\mid\mathfrak{F}_t\right]<\infty$ almost surely, i.e.

$$\mathbb{P}\left[\mathcal{L}(A(t,Y,Z)\mid\mathfrak{F}_t)\notin\mathcal{P}_2(E)\right]=0.\tag{7}$$

We now construct a progressive version of $(\mathcal{L}(A(t,Y,Z) \mid \mathfrak{F}_t))_{t\in[0,T]}$. Since $A(Y,Z)$ is an equivalence class, we fix a (measurable) representative. The resulting construction is independent of this choice up to $dt \otimes d\mathbb{P}$-a.e. equality.

As E is a Polish space, the space $\mathcal{P}_2(E)$ equipped with the 2-Wasserstein topology is also a Polish space (Bogachev, 2007, Theorems 8.3.2 and 8.9.4). Moreover, one can choose a countable family $\{f_n\}_{n\geq 1}$ of continuous bounded real functions on E that is convergence-determining, hence separates measures in $\mathcal{P}_2(E)$ (Bain and Crisan, 2009, Theorem 2.18); that is, $\nu = \nu'$ if and only if $\int_E f_n \, d\nu = \int_E f_n \, d\nu'$ for all $n \geq 1$.

Consider the injective map $\mathcal{I} : \mathcal{P}_2(E) \to \mathbb{R}^{\mathbb{N}}$ (endowed with its Borel product topology) defined by $\mathcal{I}(\nu) = \left(\int_E f_n \, d\nu\right)_{n\geq 1}$. As each f_n is bounded and continuous, $\mathcal{I}$ is continuous and thus Borel measurable. By the Lusin-Souslin Theorem on injective Borel maps (Kechris, 1995, Theorem 15.1), the image $\mathcal{P}' := \mathcal{I}(\mathcal{P}_2(E))$ is a Borel subset of $\mathbb{R}^{\mathbb{N}}$, and the inverse map $\mathcal{I}^{-1} : \mathcal{P}' \to \mathcal{P}_2(E)$ is Borel measurable.

For each $n \geq 1$, the real valued process $(f_n(A(t,Y,Z)))_{t\in[0,T]}$ is bounded and jointly measurable. By the optional projection theorem (Bass, 2011, Theorem 16.17), it admits an optional projection $\widetilde{\nu}^n$. In particular, for every $t \in [0,T]$,

$$\widetilde{\nu}^n_t = \mathbb{E}[f_n(A(t,Y,Z)) \mid \mathcal{F}_t] = \int_E f_n d\mathcal{L}(A(t,Y,Z) \mid \mathcal{F}_t) \quad \text{a.s.}$$

We consider the $\mathbb{R}^{\mathbb{N}}$ valued process $\widetilde{\nu} = (\widetilde{\nu}^n)_{n\geq 1}$. From the above identity, by definition of $\mathcal{I}$, we have for any $t \in [0,T]$, $\widetilde{\nu}_t = (\int_E f_n d\mathcal{L}(A(t,Y,Z) \mid \mathcal{F}_t))_{n\geq 1} = \mathcal{I}(\mathcal{L}(A(t,Y,Z) \mid \mathcal{F}_t))$ almost surely, and from (7), for almost every $t \in [0,T]$, we have $\widetilde{\nu}_t \in \mathcal{I}(\mathcal{P}_2(E)) = \mathcal{P}'$ almost surely. In addition, since the family $(f_n)_{n\geq 1}$ is countable, $\widetilde{\nu}$ is optional.

Let $\mathcal{N}_2 = \{(\omega,t) \in \Omega \times [0,T] : \widetilde{\nu}_t(\omega) \notin \mathcal{P}'\}$. This set is optional as the preimage of the Borel set $\mathbb{R}^{\mathbb{N}} \setminus \mathcal{P}'$ by the optional process $\widetilde{\nu}$. Moreover, one has by Fubini's theorem

$$(\mathbb{P} \otimes dt)(\mathcal{N}_2) = (\mathbb{P} \otimes dt)(\widetilde{\nu} \notin \mathcal{P}') = \int_0^T \mathbb{P}(\widetilde{\nu}_t \notin \mathcal{P}')dt = 0.$$

We now define, for any fixed $e_0 \in E$,

$$\nu_t = \widetilde{\nu}_t \mathbf{1}_{\mathcal{N}_2^c} + \mathcal{I}(\delta_{e_0})\mathbf{1}_{\mathcal{N}_2},$$

where $e_0 \in E$ is fixed. Then ν is optional, takes values in $\mathcal{P}'$, and, by the previous computations, agrees $dt \otimes d\mathbb{P}$-a.e. with $\widetilde{\nu}_t = \mathcal{I}(\mathcal{L}(A(t,Y,Z) \mid \mathfrak{F}_t))$. Therefore

$$m_t(Y,Z) := \mathcal{I}^{-1}(\nu_t)$$

is $\mathcal{P}_2(E)$ valued and optional (hence progressive), and satisfies (6) as $\mathcal{I}^{-1}(\nu_t) = \mathcal{I}^{-1}(\widetilde{\nu}_t) = \mathcal{L}(A(t,Y,Z) \mid \mathfrak{F}_t)$ almost everywhere.

Last, (4) is an immediate consequence of (5), as

$$W_2(m_t(Y,Z), m_t(Y',Z'))^2 \leq \mathbb{E}_t\left[d_E(A(t,Y,Z), A(t,Y',Z'))^2\right] \quad dt \otimes d\mathbb{P} \text{ a.e.},$$

where the conditional law of $(A(t,Y,Z), A(t,Y',Z'))$ provides the coupling in the Wasserstein bound. Finally, by the triangle and Cauchy–Schwarz inequalities and Assumption 2.1(ii),

$$\begin{aligned}\mathbb{E}\left[\int_0^T |f(s,Y_s,Z_s,m_s(Y,Z))|ds\right] \leq & \sqrt{T}\|f(\cdot,0,0,m_\cdot(0,0))\|_{\mathcal{H}^2(T)} \\ & + \Lambda\sqrt{T}\left(\|Y\|_{\mathcal{H}^2(T)} + \|Z\|_{\mathcal{H}^2(T)}\right) \\ & + \Lambda\sqrt{T}\mathbb{E}\left[\int_0^T W_2(m_s(Y,Z), m_s(0,0))^2 ds\right]^{\frac{1}{2}}.\end{aligned}$$

Using (4), Fubini's theorem, and (3), we obtain

$$\mathbb{E}\left[\int_0^T W_2(m_s(Y,Z),m_s(0,0))^2 ds\right] \le \mathbb{E}\left[\int_0^T \int_s^{T+\theta} \left(|Y_u|^2+|Z_u|^2\right)\rho_s(du)ds\right]$$
$$\le \Lambda\mathbb{E}\left[\int_0^{T+\theta}\left(|Y_u|^2+|Z_u|^2\right)\,du\right] < \infty.$$

This proves that $\int_0^T |f(s,Y_s,Z_s,m_s(Y,Z))|ds$ is almost surely finite.

Example 2.4. *(i) Classical anticipated BSDEs (Peng and Yang (2009)):* $E=\mathbb{R}^n\times\mathbb{R}^{n\times q}$, $A(t,Y,Z)=(Y_{t+\theta},Z_{t+\theta})$ for some $\theta\ge 0$, hence $\widetilde{m}_t(Y,Z):=\mathcal{L}_t(Y_{t+\theta},Z_{t+\theta})$, and $f(t,y,z,\mu)=\widetilde{f}(t,y,z,\int y\mu(dy,dz),\int z\mu(dy,dz))$. In this case, the anticipated term depends on the conditional distribution of the solution at a single future time $t+\theta$. This is the original ABSDE framework of Peng and Yang (2009), where the driver depends on $\mathbb{E}_t[Y_{t+\theta}]$ and $\mathbb{E}_t[Z_{t+\theta}]$.

Lemma 2.1 provides the progressive process $m(Y,Z)$ satisfying (6). Regarding (5), for any $(Y,Z),(Y',Z')\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})$, for dt-almost every $0\le t\le T$, we almost surely have

$$\mathbb{E}_t\left[d_{\mathbb{R}^n\times\mathbb{R}^{n\times q}}((Y_{t+\theta},Z_{t+\theta}),(Y'_{t+\theta},Z'_{t+\theta}))^2\right]=\mathbb{E}_t\left[\left(\left|Y_{t+\theta}-Y'_{t+\theta}\right|^2+\left|Z_{t+\theta}-Z'_{t+\theta}\right|^2\right)\right]$$
$$=\mathbb{E}_t\left[\int_t^{T+\theta}\left(\left|Y_s-Y'_s\right|^2+\left|Z_s-Z'_s\right|^2\right)\rho_t(ds)\right],$$

where $\rho_t(ds)=\delta_{t+\theta}(ds)$ is the Dirac measure at $t+\theta$, with mass 1.
(ii) Expected shortfall and FVA, KVA application: $n=2$, $E=\mathbb{R}$, $A(t,Y,Z)=\int_t^{t+\theta}Z_s dW_s$, hence $\widetilde{m}_t(Y,Z):=\mathcal{L}_t(\int_t^{t+\theta}Z_s dW_s)$, and $f(t,y,z,\mu)=\widetilde{f}(t,y,z,\mathbb{ES}_\alpha(\mu))$ for some $\alpha\in(0,1)$, where $\mathbb{ES}_\alpha(\mu)=\frac{\int_{\mathbb{V}a\mathbb{R}_\alpha(\mu)}^\infty x\mu(dx)}{\int_{\mathbb{V}a\mathbb{R}_\alpha(\mu)}^\infty \mu(dx)}$ is the Expected Shortfall at level α of μ (with assumingly continuous and increasing cdf), and $\mathbb{V}a\mathbb{R}_\alpha(\mu)=\inf\{x\in\mathbb{R}:\mu((-\infty,x])\ge\alpha\}$ is the quantile of μ at level α. The capital at risk of an investment bank can be modeled as an expected shortfall of its future trading gains/losses, with law μ. The variation margin is a tracker of the mark-to-market of a bank's derivative portfolio, needing to be posted by the bank as collateral to its clients. As explained in Crépey (2026, Chapter VII), the fungibility between variation margin and capital at risk leads to an ABSDE of the above form (extended to a jump-diffusion setup encompassing the defaults of the clients of the bank) for the bank's FVA and KVA, i.e. the costs for the bank of funding its variation margin and of remunerating its shareholders for their capital at risk.

Lemma 2.1 provides the progressive process $m(Y,Z)$ satisfying (6). Regarding (5), for any $Z,Z'\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^{n\times q})$, for dt-almost every $0\le t\le T$, we almost surely have

$$\mathbb{E}_t\left[d_{\mathbb{R}}(\int_t^{t+\theta}Z_s dW_s,\int_t^{t+\theta}Z'_s dW_s)^2=\right]\mathbb{E}_t\left[\left|\int_t^{t+\theta}Z_s dW_s-\int_t^{t+\theta}Z'_s dW_s\right|^2\right]$$
$$=\mathbb{E}_t\left[\int_t^{t+\theta}\left|Z_s-Z'_s\right|^2 ds\right]=\mathbb{E}_t\left[\int_t^{T+\theta}\left|Z_s-Z'_s\right|^2\rho_t(ds)\right],$$

where $\rho_t(ds)$ is the Lebesgue measure on $[t,t+\theta]$, with mass θ.
(iii) Expected shortfall and MVA application: The initial margin is an additional, segregated layer of collateral, posted by the bank as a guarantee against the gap risk of insufficient variation margin. Under the stylized specification considered in , Section 3.1AgarwalDeMarcoGobetLopez-NoubiagainZhou18 the valuation of the claim net of the corresponding MVA, i.e. the cost for the bank of funding its initial margin, satisfies an ABSDE with $n=q=1, E=\mathbb{R}, A(t,Y,Z)=Y_{t+\theta}-Y_t$ for some $\theta\ge 0$, hence $\widetilde{m}_t(Y,Z)=\mathcal{L}_t(Y_{t+\theta}-Y_t)$, and $f(t,y,z,\mu)=\widetilde{f}(t,y,z,\mathbb{ES}_\alpha(\mu))$.

Lemma 2.1 provides the progressive process $m(Y,Z)$ satisfying (6). Regarding (5), for any

$Y, Y' \in \mathcal{H}(\mathfrak{F}, \mathbb{R})$, for dt-almost every $0 \le t \le T$, we almost surely have

$$\begin{aligned}\mathbb{E}_t\left[d_{\mathbb{R}}(Y_{t+\theta} - Y_t, Y'_{t+\theta} - Y'_t)^2\right] &\le 2\mathbb{E}_t\left[\left(\left|Y_{t+\theta} - Y'_{t+\theta}\right|^2 + \left|Y_t - Y'_t\right|^2\right)\right] \\ &= \mathbb{E}_t\left[\int_t^{T+\theta} \left|Y_s - Y'_s\right|^2 \rho_t(ds)\right],\end{aligned}$$

where $\rho_t(ds) = 2\delta_t(ds) + 2\delta_{t+\theta}(ds)$, with mass 4.

Remark 2.5. Note that (4) is a Lipschitz-like assumption on m, the Lipschitz constant being hidden in ρ_t.

2.1 A Priori Estimates

Lemma 2.2. *Let f and m satisfy Assumptions 2.1(ii)-(iii). Then, for any $\beta \ge 0$ and any $(Y, Z), (Y', Z'), (\widetilde{Y}, \widetilde{Z}), (\widetilde{Y}', \widetilde{Z}') \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$, we have*

$$\begin{aligned}&\|f(\cdot, Y_\cdot, Z_\cdot, m_\cdot(\widetilde{Y}, \widetilde{Z})) - f(\cdot, Y'_\cdot, Z'_\cdot, m_\cdot(\widetilde{Y}', \widetilde{Z}'))\|^2_{\mathcal{H}^2_\beta(T)} \\ &\le 3\Lambda^2\left(\|Y - Y'\|^2_{\mathcal{H}^2_\beta(T)} + \|Z - Z'\|^2_{\mathcal{H}^2_\beta(T)} + \Lambda\|\widetilde{Y} - \widetilde{Y}'\|^2_{\mathcal{H}^2_\beta} + \Lambda\|\widetilde{Z} - \widetilde{Z}'\|^2_{\mathcal{H}^2_\beta}\right).\end{aligned}$$

In particular,

$$\begin{aligned}&\|f(\cdot, Y_\cdot, Z_\cdot, m_Z(Y)) - f(\cdot, Y'_\cdot, Z'_\cdot, m_\cdot(Y', Z'))\|^2_{\mathcal{H}^2_\beta(T)} \\ &\le 3(\Lambda+1)\Lambda^2\left(\|Y - Y'\|^2_{\mathcal{H}^2_\beta} + \|Z - Z'\|^2_{\mathcal{H}^2_\beta}\right).\end{aligned}$$

Proof. Using Assumption 2.1(ii)-(iii), we compute

$$\begin{aligned}&\mathbb{E}\left[\int_0^T e^{\beta t}\left|f(t, Y_t, Z_t, m_t(\widetilde{Y}, \widetilde{Z})) - f(t, Y'_t, Z'_t, m_t(\widetilde{Y}', \widetilde{Z}'))\right|^2 dt\right] \\ &\qquad \le 3\Lambda^2\mathbb{E}\left[\int_0^T e^{\beta t}\left(|Y_t - Y'_t|^2 + |Z_t - Z'_t|^2 + W_2(m_t(\widetilde{Y}, \widetilde{Z}), m_t(\widetilde{Y}', \widetilde{Z}'))^2\right) dt\right] \\ &= 3\Lambda^2\left(\|Y - Y'\|^2_{\mathcal{H}^2_\beta(T)} + \|Z - Z'\|^2_{\mathcal{H}^2_\beta(T)} + \mathbb{E}\left[\int_0^T e^{\beta t} W_2(m_t(\widetilde{Y}, \widetilde{Z}), m_t(\widetilde{Y}', \widetilde{Z}'))^2 dt\right]\right) \\ &\le 3\Lambda^2\left(\|Y - Y'\|^2_{\mathcal{H}^2_\beta(T)} + \|Z - Z'\|^2_{\mathcal{H}^2_\beta(T)} + \right. \\ &\quad \left.\mathbb{E}\left[\int_0^T e^{\beta t}\mathbb{E}_t\left[\int_t^{T+\theta}\left(\left|\widetilde{Y}_s - \widetilde{Y}'_s\right|^2 + \left|\widetilde{Z}_s - \widetilde{Z}'_s\right|^2\right)\rho_t(ds)\right] dt\right]\right) \\ &\le 3\Lambda^2\left(\|Y - Y'\|^2_{\mathcal{H}^2_\beta(T)} + \|Z - Z'\|^2_{\mathcal{H}^2_\beta(T)} + \right. \\ &\quad \left.\mathbb{E}\left[\int_0^T \mathbb{E}_t\left[\int_t^{T+\theta} e^{\beta s}\left(\left|\widetilde{Y}_s - \widetilde{Y}'_s\right|^2 + \left|\widetilde{Z}_s - \widetilde{Z}'_s\right|^2\right)\rho_t(ds)\right] dt\right]\right) \\ &\le 3\Lambda^2\left(\|Y - Y'\|^2_{\mathcal{H}^2_\beta(T)} + \|Z - Z'\|^2_{\mathcal{H}^2_\beta(T)} + \Lambda\|\widetilde{Y} - \widetilde{Y}'\|^2_{\mathcal{H}^2_\beta} + \Lambda\|\widetilde{Z} - \widetilde{Z}'\|^2_{\mathcal{H}^2_\beta}\right),\end{aligned} \tag{8}$$

where we used (3) for the last inequality. □

Proposition 2.3. *For (f, Φ, Ψ, m) satisfying Assumption 2.1, let (Y, Z) be an $\mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n) \times \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^{n\times q})$ solution to the ABSDE (2). Then $Y \in \mathcal{S}^2(\mathfrak{F}, \mathbb{R}^n)$.*

Proof. By (2) and the triangular inequality, we have

$$\sup_{0\le t\le T}|Y_t|^2 \le 3|\Phi_T|^2 + 3T\int_0^T |f(t, Y_t, Z_t, m_t(Y, Z))|^2 dt + 3\sup_{0\le t\le T}\left|\int_t^T Z_s dW_s\right|^2.$$

First, by Assumption 2.1(i), one has $\mathbb{E}\left[|\Phi_T|^2\right] \leq \mathbb{E}\left[\sup_{T\leq s\leq T+\theta}|\Phi_s|^2\right] < \infty$.
Second, using Assumption 2.1(ii) and Lemma 2.2 with $\beta = 0$, we compute

$$\begin{aligned}
&\mathbb{E}\left[\int_0^T |f(t,Y_t,Z_t,m_t(Y,Z))|^2\,dt\right] \\
&\leq 2\mathbb{E}\left[\int_0^T |f(t,Y_t,Z_t,m_t(Y,Z)) - f(t,0,0,m_t(0,0))|^2\,dt\right] \\
&\quad + 2\mathbb{E}\left[\int_0^T |f(t,0,0,m_t(0,0))|^2\,dt\right] \\
&\leq 6(\Lambda+1)\Lambda^2\|(Y,Z)\|^2_{\mathcal{H}^2} + 2\|f(\cdot,0,0,m_\cdot(0,0))\|^2_{\mathcal{H}^2(T)}.
\end{aligned} \tag{9}$$

Third, the Burkholder-Davis-Gundy inequality of Karatzas and Shreve (1991, Theorem 3.28) yields

$$\begin{aligned}
\mathbb{E}\left[\sup_{0\leq t\leq T}\left|\int_t^T Z_s dW_s\right|^2\right] &\leq 2\mathbb{E}\left[\left|\int_0^T Z_s dW_s\right|^2\right] + 2\mathbb{E}\left[\sup_{0\leq s\leq T}\left|\int_0^s Z_r dW_r\right|^2\right] \\
&\leq 10\mathbb{E}\left[\int_0^T |Z_s|^2 ds\right] < \infty.
\end{aligned}$$

All in all, this proves that $\|Y\|^2_{\mathcal{S}^2(T)} < \infty$.
Last, $\|Y\|^2_{\mathcal{S}^2} \leq \|Y\|^2_{\mathcal{S}^2(T)} + \|\Phi\|^2_{\mathcal{S}^2(T,T+\theta)} < \infty$, by Assumption 2.1(i), which proves that $Y \in \mathcal{S}^2(\mathfrak{F},\mathbb{R}^n)$, as Y is continuous by (2). □

Proposition 2.4. *For $i = 1,2$, let (Y^i, Z^i) be an $\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n) \times \mathcal{H}^2(\mathfrak{F},\mathbb{R}^{n\times q})$ solution to the AB-SDE (2) with parameters $(f^i, \Phi^i, \Psi^i, m^i)$ satisfying Assumption 2.1 for some Lipschitz constant Λ^i of f^i and with associated admissible kernel ρ^i in Assumption 2.1(iii), whose admissibility constant is bounded by Λ^i. We set $m^i_t = m^i_t(Y^i, Z^i)$, $i = 1,2$, and $m^{1,2}_t = m^1_t(Y^2, Z^2)$, $t \in [0,T]$ and we define,*

$$\begin{aligned}
&\delta Y_t = Y^1_t - Y^2_t, \delta Z_t = Z^1_t - Z^2_t, \quad t \in [0, T+\theta], \\
&\delta\Phi_t = \Phi^1_t - \Phi^2_t, \delta\Psi_t = \Psi^1_t - \Psi^2_t, \quad t \in [T, T+\theta], \text{ and} \\
&\delta m_t = W_2(m^1_t, m^{1,2}_t), \delta_2 f_t = f^1(t, Y^2_t, Z^2_t, m^{1,2}_t) - f^2(t, Y^2_t, Z^2_t, m^2_t), \quad t \in [0,T].
\end{aligned}$$

Then, for any $\nu, \lambda, \eta, \beta > 0$ such that

$$\Lambda^1(2 + \lambda^2 + \nu^2 + \frac{\Lambda^1}{\nu^2}) + \eta^2 \leq \beta \text{ and } \Lambda^1(\frac{1}{\lambda^2} + \frac{\Lambda^1}{\nu^2}) < 1, \tag{10}$$

we have

$$\|\delta Y\|^2_{\mathcal{H}^2_\beta(T)} \leq T(1+\theta)\|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + \frac{T}{\eta^2}\|\delta_2 f\|^2_{\mathcal{H}^2_\beta(T)} + T\|\delta\Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)} \tag{11}$$

$$\|\delta Z\|^2_{\mathcal{H}^2_\beta(T)} \leq \tag{12}$$

$$\frac{\nu^2\lambda^2}{\nu^2\lambda^2 - \Lambda^1(\nu^2 + \Lambda^1\lambda^2)}\left((1+\theta)\|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + \frac{1}{\eta^2}\|\delta_2 f\|^2_{\mathcal{H}^2_\beta(T)} + \|\delta\Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)}\right).$$

In addition, for fixed $\lambda, \nu, \eta, \beta$ chosen once and for all satisfying (10), there exists a real constant C only depending on (T, θ, Λ^1) such that

$$\|\delta Y\|^2_{\mathcal{S}^2_\beta(T)} \leq C\left(\|\delta_2 f\|^2_{\mathcal{H}^2_\beta(T)} + \|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + \|\delta\Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)}\right). \tag{13}$$

Proof. For $t \in [0,T]$, applying the Itô formula to $|\delta Y_t|^2$ and the Cauchy-Schwarz inequality, we obtain

$$
\begin{aligned}
&e^{\beta t}|\delta Y_t|^2 + \beta \int_t^T e^{\beta s}|\delta Y_s|^2 ds + \int_t^T e^{\beta s}|\delta Z_s|^2 ds \\
=&e^{\beta T}|\delta Y_T|^2 + 2\int_t^T e^{\beta s}\langle \delta Y_s, f^1(s, Y_s^1, Z_s^1, m_s^1) - f^2(s, Y_s^2, Z_s^2, m_s^2)\rangle ds \\
&- 2\int_t^T e^{\beta s}\delta Y_s^\top \delta Z_s dW_s \\
\leq&e^{\beta T}|\delta \Phi_T|^2 + 2\int_t^T e^{\beta s}|\delta Y_s||f^1(s, Y_s^1, Z_s^1, m_s^1) - f^2(s, Y_s^2, Z_s^2, m_s^2)| ds \\
&- 2\int_t^T e^{\beta s}\delta Y_s^\top \delta Z_s dW_s \\
\leq&e^{\beta T}|\delta \Phi_T|^2 + 2\int_t^T e^{\beta s}|\delta Y_s||f^1(s, Y_s^1, Z_s^1, m_s^1) - f^1(s, Y_s^2, Z_s^2, m_s^{1,2})| ds \\
&+ 2\int_t^T e^{\beta s}|\delta Y_s||\delta_2 f_s| ds - 2\int_t^T e^{\beta s}\delta Y_s^\top \delta Z_s dW_s.
\end{aligned} \tag{14}
$$

We first prove (11)-(12). By Proposition 2.3, $\delta Y \in \mathcal{S}^2(\mathfrak{F}, \mathbb{R}^n)$; by assumption, $\delta Z \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^{n\times q})$; hence $(e^{\beta s}\delta Y_s^\top \delta Z_s)_{s\in[0,T]} \in \mathcal{H}^1(\mathfrak{F}, \mathbb{R}^q)$. In particular, for any $t \leq T$, $\int_t^T e^{\beta s}\delta Y_s^\top \delta Z_s dW_s$ is integrable with zero expectation. Hence

$$
\begin{aligned}
&\mathbb{E}\left[e^{\beta t}|\delta Y_t|^2\right] + \beta\|\delta Y\|^2_{\mathcal{H}^2_\beta(t,T)} + \|\delta Z\|^2_{\mathcal{H}^2_\beta(t,T)} \leq \\
&\qquad \mathbb{E}\left[e^{\beta T}|\delta \Phi_T|^2\right] + 2\mathbb{E}\left[\int_t^T e^{\beta s}|\delta Y_s||f^1(s, Y_s^1, Z_s^1, m_s^1) - f^1(s, Y_s^2, Z_s^2, m_s^{1,2})| ds\right] \\
&\qquad\qquad + 2\mathbb{E}\left[\int_t^T e^{\beta s}|\delta Y_s||\delta_2 f_s| ds\right].
\end{aligned}
$$

Using Assumption 2.1(ii) on f^1 and the Young inequality, we obtain

$$
\begin{aligned}
&\mathbb{E}\left[e^{\beta t}|\delta Y_t|^2\right] + \beta\|\delta Y\|^2_{\mathcal{H}^2_\beta(t,T)} + \|\delta Z\|^2_{\mathcal{H}^2_\beta(t,T)} \\
\leq&\|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + 2\Lambda^1\mathbb{E}\left[\int_t^T e^{\beta s}|\delta Y_s|\left(|\delta Y_s| + |\delta Z_s| + \delta m_s\right) ds\right] + 2\mathbb{E}\left[\int_t^T e^{\beta s}|\delta Y_s||\delta_2 f_s| ds\right] \\
\leq&\|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + \left(\Lambda^1\left(2 + \lambda^2 + \nu^2\right) + \eta^2\right)\|\delta Y\|^2_{\mathcal{H}^2_\beta(t,T)} \\
&+ \frac{\Lambda^1}{\lambda^2}\|\delta Z\|^2_{\mathcal{H}^2_\beta(t,T)} + \frac{\Lambda^1}{\nu^2}\|\delta m\|^2_{\mathcal{H}^2_\beta(t,T)} + \frac{1}{\eta^2}\|\delta_2 f\|^2_{\mathcal{H}^2_\beta(t,T)}.
\end{aligned} \tag{15}
$$

From Assumption 2.1(iii), using $e^{\beta s} \leq e^{\beta r}$ for $s \leq r$, we also have

$$\begin{aligned}
&\|\delta m\|^2_{\mathcal{H}^2_\beta(t,T)} = \mathbb{E}\left[\int_t^T e^{\beta s} W_2^2(m_s^1, m_s^{1,2}) ds\right] \\
\leq &\mathbb{E}\left[\int_t^T e^{\beta s} \mathbb{E}_s\left[\int_s^{T+\theta} \left(|\delta Y_r|^2 + |\delta Z_r|^2\right) \rho_s^1(dr) ds\right]\right] \\
\leq &\mathbb{E}\left[\int_t^T \int_s^{T+\theta} e^{\beta r} \left(|\delta Y_r|^2 + |\delta Z_r|^2\right) \rho_s^1(dr) ds\right] \\
\leq &\Lambda^1 \mathbb{E}\left[\int_t^{T+\theta} e^{\beta r} \left(|\delta Y_r|^2 + |\delta Z_r|^2\right) dr\right] \\
= &\Lambda^1 \mathbb{E}\left[\int_t^T e^{\beta r} \left(|\delta Y_r|^2 + |\delta Z_r|^2\right) dr + \int_T^{T+\theta} e^{\beta r} \left(|\delta \Phi_r|^2 + |\delta \Psi_r|^2\right) dr\right] \\
\leq &\Lambda^1 \|\delta Y\|^2_{\mathcal{H}^2_\beta(t,T)} + \Lambda^1 \|\delta Z\|^2_{\mathcal{H}^2_\beta(t,T)} + \Lambda^1 \theta \|\delta \Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + \Lambda^1 \|\delta \Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)}.
\end{aligned} \tag{16}$$

Combining (15) and (16), we obtain

$$\begin{aligned}
&\mathbb{E}\left[e^{\beta t}|\delta Y_t|^2\right] + \left(\beta - \Lambda^1\left(2 + \lambda^2 + \nu^2 + \frac{\Lambda^1}{\nu^2}\right) - \eta^2\right)\|\delta Y\|^2_{\mathcal{H}^2_\beta(t,T)} \\
&\qquad + \left(1 - \Lambda^1\left(\frac{1}{\lambda^2} + \frac{\Lambda^1}{\nu^2}\right)\right)\|\delta Z\|^2_{\mathcal{H}^2_\beta(t,T)} \\
&\leq \left(1 + \frac{(\Lambda^1)^2\theta}{\nu^2}\right)\|\delta \Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + \frac{(\Lambda^1)^2}{\nu^2}\|\delta \Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)} + \frac{1}{\eta^2}\|\delta_2 f\|^2_{\mathcal{H}^2_\beta(T)}.
\end{aligned} \tag{17}$$

This implies, for $(\lambda, \eta, \nu, \beta)$ satisfying (10),

$$\mathbb{E}\left[e^{\beta t}|\delta Y_t|^2\right] \leq (1+\theta)\|\delta \Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + \frac{1}{\eta^2}\|\delta_2 f\|^2_{\mathcal{H}^2_\beta(T)} + \|\delta \Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)},$$

as in particular $\frac{(\Lambda^1)^2}{\nu^2} < 1$. Integrating over time, we obtain (11). Using (10) again, (17) (at $t = 0$) implies (12).

To prove (13), starting from (14), we obtain

$$\begin{aligned}
e^{\beta t}|\delta Y_t|^2 \leq\ & e^{\beta T}|\delta \Phi_T|^2 + 2\int_0^T e^{\beta s}|\delta Y_s|^2 ds + \int_0^T e^{\beta s}|f^1(s, Y_s^1, Z_s^1, m_s^1) - f^1(s, Y_s^2, Z_s^2, m_s^{1,2})|^2 ds \\
&+ \int_0^T e^{\beta s}|\delta_2 f_s|^2 ds + 2\sup_{0\leq s\leq T}\left|\int_s^T e^{\beta r} \delta Y_r^\top \delta Z_r dW_r\right|.
\end{aligned}$$

As the right hand side does not depend on t, taking the supremum over t and expectations yields

$$\begin{aligned}
\|\delta Y\|^2_{\mathcal{S}^2_\beta(T)} \leq\ & \|\delta \Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + 2\|\delta Y\|^2_{\mathcal{H}^2_\beta(T)} \\
&+ \mathbb{E}\left[\int_0^T e^{\beta s}|f^1(s, Y_s^1, Z_s^1, m_s^1) - f^1(s, Y_s^2, Z_s^2, m_s^{1,2})|^2 ds\right] \\
&+ \|\delta_2 f\|^2_{\mathcal{H}^2_\beta(T)} + 2\mathbb{E}\left[\sup_{0\leq s\leq T}\left|\int_s^T e^{\beta r} \delta Y_r^\top \delta Z_r dW_r\right|\right].
\end{aligned}$$

From Lemma 2.2, as $\|\delta Y\|^2_{\mathcal{H}^2_\beta} \leq \|\delta Y\|^2_{\mathcal{H}^2_\beta(T)} + \theta\|\delta \Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)}$ and $\|\delta Z\|^2_{\mathcal{H}^2_\beta} = \|\delta Z\|^2_{\mathcal{H}^2_\beta(T)} +$

$\|\delta\Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)}$, we obtain

$$\begin{aligned}\|\delta Y\|^2_{\mathcal{S}^2_\beta(T)} &\leq (1+3\theta(\Lambda^1+1)(\Lambda^1)^2)\|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + 3(\Lambda^1+1)(\Lambda^1)^2\|\delta\Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)} \\ &\quad + (2+3(\Lambda^1+1)(\Lambda^1)^2)\|\delta Y\|^2_{\mathcal{H}^2_\beta(T)} + 3(\Lambda^1+1)(\Lambda^1)^2\|\delta Z\|^2_{\mathcal{H}^2_\beta(T)} \\ &\quad + \|\delta_2 f\|^2_{\mathcal{H}^2_\beta(T)} + 2\mathbb{E}\left[\sup_{0\leq s\leq T}\left|\int_s^T e^{\beta r}\delta Y_r^\top \delta Z_r dW_r\right|\right].\end{aligned} \tag{18}$$

Moreover, the Burkholder-Davis-Gundy and Young inequalities imply the existence of some $C>0$ such that

$$\begin{aligned}&\mathbb{E}\left[\sup_{0\leq s\leq T}\left|\int_s^T e^{\beta r}\delta Y_r^\top \delta Z_r dW_r\right|\right] \leq \mathbb{E}\left[\left|\int_0^T e^{\beta r}\delta Y_r^\top \delta Z_r dW_r\right|\right] \\ &\qquad + \mathbb{E}\left[\sup_{0\leq s\leq T}\left|\int_0^s e^{\beta r}\delta Y_r^\top \delta Z_r dW_r\right|\right] \\ &\leq 2\mathbb{E}\left[\sup_{0\leq s\leq T}\left|\int_0^s e^{\beta r}\delta Y_r^\top \delta Z_r dW_r\right|\right] \leq 2C\mathbb{E}\left[\left(\int_0^T e^{2\beta r}|\delta Y_r|^2|\delta Z_r|^2 dr\right)^{\frac{1}{2}}\right] \\ &\leq 2C\mathbb{E}\left[\left(\sup_{0\leq r\leq T} e^{\beta r}|\delta Y_r|^2\right)^{\frac{1}{2}}\left(\int_0^T e^{\beta r}|\delta Z_r|^2 dr\right)^{\frac{1}{2}}\right] \leq \frac{1}{4}\|\delta Y\|^2_{\mathcal{S}^2_\beta(T)} + 4C^2\|\delta Z\|^2_{\mathcal{H}^2_\beta(T)}.\end{aligned} \tag{19}$$

All in all, plugging (19) into (18) and using (11)-(12), we obtain (13). □

From these a priori estimates, we immediately deduce some regularity properties.

Corollary 2.5. *Let $(f^\alpha,\Phi^\alpha,\Psi^\alpha,m^\alpha,\alpha\in\mathbb{R})$ be a family of parameters satisfying Assumption 2.1 with Λ independent of α. Assume that, for each $\alpha\in\mathbb{R}$, there exists an $\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F},\mathbb{R}^{n\times q})$ solution (Y^α,Z^α) to the ABSDE (2) with parameters $(f^\alpha,\Phi^\alpha,\Psi^\alpha,m^\alpha)$. Assume in addition that, as $\alpha\to\alpha_0\in\mathbb{R}$, we have*

$$\begin{aligned}&\left\|f^\alpha\left(\cdot,Y^{\alpha_0},Z^{\alpha_0},m^\alpha_\cdot(Y^{\alpha_0},Z^{\alpha_0})\right) - f^{\alpha_0}\left(\cdot,Y^{\alpha_0},Z^{\alpha_0},m^{\alpha_0}_\cdot(Y^{\alpha_0},Z^{\alpha_0})\right)\right\|_{\mathcal{H}^2(T)}\to 0,\\ &\|\Phi^\alpha-\Phi^{\alpha_0}\|_{\mathcal{S}^2(T,T+\theta)}\to 0,\ \text{and}\ \|\Psi^\alpha-\Psi^{\alpha_0}\|_{\mathcal{H}^2(T,T+\theta)}\to 0.\end{aligned}$$

Then the function $\alpha\to(Y^\alpha,Z^\alpha)$ from $\mathbb{R}$ to $\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F},\mathbb{R}^{n\times q})$ is continuous at α_0.

Proof. We have

$$\begin{aligned}\|Y^\alpha-Y^{\alpha_0}\|^2_{\mathcal{H}^2} &\leq \|Y^\alpha-Y^{\alpha_0}\|^2_{\mathcal{H}^2(T)} + \theta\|\Phi^\alpha-\Phi^{\alpha_0}\|^2_{\mathcal{S}^2(T,T+\theta)},\\ \|Z^\alpha-Z^{\alpha_0}\|^2_{\mathcal{H}^2} &= \|Z^\alpha-Z^{\alpha_0}\|^2_{\mathcal{H}^2(T)} + \|\Psi^\alpha-\Psi^{\alpha_0}\|^2_{\mathcal{H}^2(T,T+\theta)}.\end{aligned}$$

The second terms on the right-hand sides converge to 0 by assumption, while the first terms converge to 0 as an immediate consequence of the a priori estimates (11)–(12) applied with $(f^1,\Phi^1,\Psi^1,m^1)=(f^\alpha,\Phi^\alpha,\Psi^\alpha,m^\alpha)$ and $(f^2,\Phi^2,\Psi^2,m^2)=(f^{\alpha_0},\Phi^{\alpha_0},\Psi^{\alpha_0},m^{\alpha_0})$, using the equivalence of the $\mathcal{H}^2$ (resp. $\mathcal{S}^2$) and $\mathcal{H}^2_\beta$ (resp. $\mathcal{S}^2_\beta$) norms and noting that $\delta_2 f$, $\delta\Phi$ and $\delta\Psi$ all converge to 0 as $\alpha\to\alpha_0$ by assumption. □

Note that the function from $\mathbb{R}$ to $\mathcal{S}^2(\mathfrak{F},\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F},\mathbb{R}^{n\times q})$ is continuous at α_0 too, as we have proved that $\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F},\mathbb{R}^{n\times q})$ solutions actually belong to $\mathcal{S}^2(\mathfrak{F},\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F},\mathbb{R}^{n\times q})$, and (12)-(13) controls their differences in $\mathcal{S}^2(\mathfrak{F},\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F},\mathbb{R}^{n\times q})$.

2.2 Existence and Uniqueness

Lemma 2.6. *Let (f, Φ, Ψ, m) satisfy Assumption 2.1. For any $(\widetilde{Y}, \widetilde{Z}) \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$, we consider the classical BSDE with terminal condition Φ_T and driver $(\omega, t, y, z) \mapsto f(\omega, t, y, z, m_t(\widetilde{Y}, \widetilde{Z})(\omega))$. This classical BSDE admits a unique $\mathcal{H}^2(\mathfrak{F}^T, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ solution.*

Proof. By definition, the driver $(\omega, t, y, z) \mapsto f(\omega, t, y, z, m_t(\widetilde{Y}, \widetilde{Z})(\omega))$ is $Prog\otimes\mathcal{B}(\mathbb{R}^n)\otimes\mathcal{B}(\mathbb{R}^{n\times q})$ measurable.
Under Assumption 2.1, we have $\mathbb{E}\left[|\Phi_T|^2\right] \leq \|\Phi\|^2_{\mathcal{S}^2} < \infty$, for $(\omega, t, y, y', z, z') \in \Omega \times [0,T] \times (\mathbb{R}^n)^2 \times (\mathbb{R}^{n\times q})^2$, $|f(\omega, t, y, z, m_t(\widetilde{Y}, \widetilde{Z})(\omega)) - f(\omega, t, y', z', m_t(\widetilde{Y}, \widetilde{Z})(\omega))| \leq \Lambda(|y - y'| + |z - z'|)$, and,

$$
\begin{aligned}
&\|f(t, 0, 0, m_t(\widetilde{Y}, \widetilde{Z}))\|^2_{\mathcal{H}^2(T)} \\
&\quad\leq 2\|f(\cdot, 0, 0, m_\cdot(\widetilde{Y}, \widetilde{Z})) - f(\cdot, 0, 0, m_\cdot(0, 0))\|^2_{\mathcal{H}^2(T)} + 2\|f(\cdot, 0, 0, m_\cdot(0, 0))\|^2_{\mathcal{H}^2(T)} \\
&\quad\leq 2\Lambda^2\|W_2(m_\cdot(\widetilde{Y}, \widetilde{Z}), m_\cdot(0, 0))\|^2_{\mathcal{H}^2(T)} + 2\|f(\cdot, 0, 0, m_\cdot(0, 0))\|^2_{\mathcal{H}^2(T)} \\
&\quad\leq 2\Lambda^2\mathbb{E}\left[\int_0^T \mathbb{E}_t\left[\int_t^{T+\theta} (\|\widetilde{Y}_s\|^2 + \left|\widetilde{Z}_s\right|^2)\rho_t(ds)\right] dt\right] + 2\|f(\cdot, 0, 0, m_\cdot(0, 0))\|^2_{\mathcal{H}^2(T)} \\
&\quad\leq 2\Lambda^3\|(\widetilde{Y}, \widetilde{Z})\|^2_{\mathcal{H}^2} + 2\|f(\cdot, 0, 0, m_\cdot(0, 0))\|^2_{\mathcal{H}^2(T)} < \infty.
\end{aligned}
$$

Hence the classical BSDE with terminal condition Φ_T and driver $f(\omega, t, y, z, m_t(\widetilde{Y}, \widetilde{Z})(\omega))$ admits a unique $\mathcal{H}^2(\mathfrak{F}^T, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ solution, by El Karoui et al. (1997, Theorem 2.1). □
For any $(\widetilde{Y}, \widetilde{Z}) \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$, we define $\Gamma(\widetilde{Y}, \widetilde{Z}) \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ as this unique $\mathcal{H}^2(\mathfrak{F}^T, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ solution on $[0, T]$ and as (Φ, Ψ) on $[T, T + \theta]$.
Note that, since $Y_{|[0,T]} \in \mathcal{S}^2(\mathfrak{F}^T, \mathbb{R}^n)$, Φ is continuous on $[T, T + \theta]$ and satisfies

$$
\mathbb{E}\left[\sup_{T\leq t\leq T+\theta} |\Phi_t|^2\right] < \infty,
$$

and $(Y_{|[0,T]})_T = \Phi_T$, the first component of $\Gamma(\widetilde{Y}, \widetilde{Z})$ is in $\mathcal{S}^2(\mathfrak{F}, \mathbb{R}^n)$.

The main result of this section is:

Theorem 2.7. *Let (f, Φ, Ψ, m) satisfy Assumption 2.1.*
(i) *The ABSDE (2) admits a unique $\mathcal{S}^2(\mathfrak{F}, \mathbb{R}^n) \times \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^{n\times q})$ solution (Y, Z), which can be characterized as the unique fixed point of the contraction Γ of $\mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$.*
(ii) *The sequence of Picard iterates defined by any $(Y^0, Z^0) \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ and, for $k \geq 0$, $(Y^{k+1}, Z^{k+1}) = \Gamma(Y^k, Z^k)$, converges to (Y, Z) in $\mathcal{H}^2(\mathfrak{F}^T, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ and $d\mathbb{P} \otimes ds$ almost everywhere. The sequence $(Y^k)_{k\geq 1}$ also converges to Y in $\mathcal{S}^2(\mathfrak{F}^T, \mathbb{R}^n)$.*

Proof. (i) Note that a pair (Y, Z) is a fixed point of Γ if and only if it solves the ABSDE. We prove that Γ is a contraction of $(\mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q}), \|\cdot\|_\beta)$, for β large enough, which will be enough to obtain the existence and uniqueness of an $\mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ solution to (2). Then, by Proposition 2.3, we obtain that $(Y, Z) \in \mathcal{S}^2(\mathfrak{F}, \mathbb{R}^n) \times \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^{n\times q})$, proving (i).

For $(\widetilde{Y}, \widetilde{Z}), (\widetilde{Y}', \widetilde{Z}') \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$, we set $(Y, Z) = \Gamma(\widetilde{Y}, \widetilde{Z})$, $(Y', Z') = \Gamma(\widetilde{Y}', \widetilde{Z}')$, $\delta Y = Y - Y'$, $\delta Z = Z - Z'$, $\delta\widetilde{Y} = \widetilde{Y} - \widetilde{Y}'$ and $\delta\widetilde{Z} = \widetilde{Z} - \widetilde{Z}'$. Note that $\delta Y = \delta Z = 0$ on $(T, T+\theta]$, so that in particular $\|(\delta Y, \delta Z)\|_{\mathcal{H}^2_\beta} = \|(\delta Y, \delta Z)\|_{\mathcal{H}^2_\beta(T)}$. By the Itô formula, the Cauchy-Schwarz inequality, and the Young inequality, we have, as also $Y_T = Y'_T = \Phi_T$ (cf. the beginning of the

proof of Proposition 2.4),

$$
\begin{aligned}
&|\delta Y_0|^2 + \int_0^T e^{\beta s}\left(\beta|\delta Y_s|^2 + |\delta Z_s|^2\right) ds \\
&= 2\int_0^T e^{\beta s}\delta Y_s^\top \big(f(s, Y_s, Z_s, m_s(\widetilde{Y}, \widetilde{Z})) - f(s, Y'_s, Z'_s, m_s(\widetilde{Y}', \widetilde{Z}'))\big) ds \\
&\qquad - 2\int_0^T e^{\beta s}\delta Y_s^\top \delta Z_s dW_s \\
&\leq \int_0^T e^{\beta s}(\frac{\beta}{2}|\delta Y_s|^2 + \frac{2}{\beta}|f(s, Y_s, Z_s, m_s(\widetilde{Y}, \widetilde{Z})) - f(s, Y'_s, Z'_s, m_s(\widetilde{Y}', \widetilde{Z}'))|^2) ds \\
&\qquad - 2\int_0^T e^{\beta s}\delta Y_s^\top \delta Z_s dW_s.
\end{aligned}
$$

Hence we obtain by Assumption 2.1, with Λ the Lipschitz constant of f,

$$
\begin{aligned}
&\mathbb{E}\left[\int_0^T e^{\beta s}(\frac{\beta}{2}|\delta Y_s|^2 + |\delta Z_s|^2) ds\right] \\
&\leq \frac{2}{\beta}\mathbb{E}\left[\int_0^T e^{\beta s}|f(s, Y_s, Z_s, m_s(\widetilde{Y}, \widetilde{Z})) - f(s, Y'_s, Z'_s, m_s(\widetilde{Y}', \widetilde{Z}'))|^2 ds\right] \\
&\leq \frac{6\Lambda^2}{\beta}\mathbb{E}\left[\int_0^T e^{\beta s}(|\delta Y_s|^2 + |\delta Z_s|^2) ds\right] + \frac{6\Lambda^3}{\beta}\mathbb{E}\left[\int_0^{T+\theta} e^{\beta s}(|\delta \widetilde{Y}_s|^2 + |\delta \widetilde{Z}_s|^2) ds\right].
\end{aligned}
$$

For $\beta > 2 \vee 6\Lambda^2 + 12\Lambda^3$, this implies

$$
\|(\delta Y, \delta Z)\|^2_{\mathcal{H}^2_\beta} = \|(\delta Y, \delta Z)\|^2_{\mathcal{H}^2_\beta(T)} \leq \frac{6\Lambda^3}{\beta - 6\Lambda^2}\|(\delta \widetilde{Y}, \delta \widetilde{Z})\|^2_{\mathcal{H}^2_\beta} \leq \frac{1}{2}\|(\delta \widetilde{Y}, \delta \widetilde{Z})\|^2_{\mathcal{H}^2_\beta}, \tag{20}
$$

which shows that Γ is a contraction of $\mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$, hence (2) admits a unique $\mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ solution.
(ii) First, from the Banach fixed-point theorem, (Y^k, Z^k) converges to (Y, Z) in $\mathcal{H}^2_\beta$. From (20), Fubini-Tonelli's theorem and the Cauchy-Schwarz inequality, we obtain

$$
\begin{aligned}
&\mathbb{E}\left[\int_0^{T+\theta} \sum_{k\geq 0}\left(|Y_t^{k+1} - Y_t^k| + |Z_t^{k+1} - Z_t^k|\right) dt\right] \\
&\qquad = \sum_{k\geq 0}\mathbb{E}\left[\int_0^{T+\theta}\left(|Y_t^{k+1} - Y_t^k| + |Z_t^{k+1} - Z_t^k|\right) dt\right] \\
&\qquad \leq \sqrt{2(T+\theta)}\sum_{k\geq 0}\|(Y^{k+1} - Y^k, Z^{k+1} - Z^k)\|_{\mathcal{H}^2_\beta} \\
&\qquad \leq \sqrt{2(T+\theta)}\|(Y^1 - Y^0, Z^1 - Z^0)\|_{\mathcal{H}^2_\beta}\sum_{k\geq 0} 2^{-\frac{k}{2}} \\
&\qquad < \infty,
\end{aligned}
$$

so that $\sum_{k\geq 0}\left(|Y^{k+1} - Y^k| + |Z^{k+1} - Z^k|\right) < \infty$ almost everywhere. Hence the sequence (Y^k, Z^k) is a Cauchy (hence convergent) sequence almost everywhere, and its limit is identified with (Y, Z) by the convergence in $\mathcal{H}^2$.

To prove the convergence of $(Y^k)_{k\geq 1}$ in $\mathcal{S}^2(\mathfrak{F}^T, \mathbb{R}^n)$, we apply the estimate (13) to the two classical BSDEs defining $(Y^{k+1}, Z^{k+1}) = \Gamma(Y^k, Z^k)$ and $(Y^k, Z^k) = \Gamma(Y^{k-1}, Z^{k-1})$. As the

terminal conditions coincide, this yields, for some constant $C \geq 0$ independent of k,

$$\begin{aligned}\|Y^{k+1} - Y^k\|^2_{\mathcal{S}^2_\beta(T)} &\leq C\|f(\cdot, Y^k, Z^k, m_\cdot(Y^k, Z^k)) - f(\cdot, Y^k, Z^k, m_\cdot(Y^{k-1}, Z^{k-1}))\|^2_{\mathcal{H}^2_\beta(T)} \\ &\leq C\Lambda^3\|(Y^k - Y^{k-1}, Z^k - Z^{k-1})\|^2_{\mathcal{H}^2_\beta} \\ &\leq C\Lambda^3 2^{-k+1}\|(Y^1 - Y^0, Z^1 - Z^0)\|^2_{\mathcal{H}^2_\beta},\end{aligned}$$

where we used Assumption 2.1 and (20). Hence $\sum_{k\geq 1}\|Y^{k+1} - Y^k\|_{\mathcal{S}^2_\beta(T)} < \infty$, so $(Y^k)_{k\geq 1}$ is Cauchy in the Banach space $\mathcal{S}^2(\mathfrak{F}^T, \mathbb{R}^n)$. Since already $(Y^k)_{k\geq 1}$ converges to Y in $\mathcal{H}^2(\mathfrak{F}^T, \mathbb{R}^n)$, this proves convergence also in $\mathcal{S}^2(\mathfrak{F}^T, \mathbb{R}^n)$. □

3 Markovian ABSDEs

The Markovian case, with all randomness in the data driven by an underlying Markovian factor process X, deserves special attention for several reasons. First, it allows representing the solution (Y, Z) of the BSDE as deterministic functions of the time and the underlying forward process X, which will be used for proving the convergence of the time-discretization scheme in Section 4. Second, this deterministic representation enables us to obtain a Feynman-Kac ABSDE representation $Y_t = u(t, X_t)$, where the value function u is connected to a nonlocal PDE through a verification argument. Finally, the Markovian framework is the typical setting for applications, where the anticipated features only depend on the current state of the underlying system and not on the past history of randomness.

The key ABSDE technical challenge in the Markovian case is to establish that the anticipated term inherits the Markov property. This requires showing that conditional distributions of functionals of future paths can be expressed as deterministic (measure valued) functions of the current state, despite their apparent dependence on future randomness.

We consider the following Markovian ABSDE setup: For $(t, x) \in [0, T] \times \mathbb{R}^d$,

$$\begin{aligned}X_s &= x + \int_t^s b(r, X_r)dr + \int_t^s \sigma(r, X_r)dW_r, \quad s \in [t, T+\theta], \\ Y_s &= \Phi(s \vee T, X_{s\vee T}) + \int_{s\wedge T}^T f(r, X_r, Y_r, Z_r, m_r(Y, Z))dr - \int_{s\wedge T}^T Z_r dW_r, \quad s \in [t, T+\theta], \\ Z_s &= \Psi(s, X_{s\vee T}), \quad s \in (T, T+\theta],\end{aligned} \tag{21}$$

for measurable maps $(b, \sigma) : [0, T+\theta] \times \mathbb{R}^d \to \mathbb{R}^d \times \mathbb{R}^{d\times q}$, $(\Phi, \Psi) : [0, T+\theta] \times \mathbb{R}^d \to \mathbb{R}^n \times \mathbb{R}^{n\times q}$, $f : [0, T] \times \mathbb{R}^d \times \mathbb{R}^n \times \mathbb{R}^{n\times q} \times \mathcal{P}_2(E) \to \mathbb{R}^n$ and $m : \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q}) \to \mathcal{H}^0(\mathfrak{F}, \mathcal{P}_2(E))$, where $d \geq 1$ is an integer.

If (X, Y, Z) solves (21), we extend the processes to $[0, t)$ by setting $X_s = x$, $Z_s = 0$, and by defining $(Y_s)_{0\leq s\leq t}$ as the unique solution of the deterministic anticipated equation

$$Y_s = Y_t + \int_s^t f(r, x, Y_r, 0, m_r(Y, Z))\, dr, \qquad 0 \leq s \leq t,$$

where existence and uniqueness of solutions to this ODE follow from the same contraction argument as in Theorem 2.7, applied on $[0, t]$ with $Z = 0$ on $[0, t]$ and (Y, Z) solving (21) on $[t, T+\theta]$ to be used for the anticipated term.

We say that the coefficients are standard Markovian parameters if they satisfy the following:

Assumption 3.1. (i) The maps Φ and Ψ are continuous, such that

$$\mathbb{E}\left[\sup_{T\le t\le T+\theta}|\Phi(t,0)|^2+\int_T^{T+\theta}|\Psi(t,0)|^2dt\right]<\infty, \tag{22}$$

and there exists a real constant Λ such that, for any $(t,x,x')\in[T,T+\theta]\times\mathbb{R}^d\times\mathbb{R}^d$,

$$\big|\Phi(t,x)-\Phi(t,x')\big|+\big|\Psi(t,x)-\Psi(t,x')\big|\le\Lambda|x-x'|.$$

Last, there exists a measurable map $\widetilde{\Psi}:[T,T+\theta]\times\mathbb{R}^d\to\mathbb{R}^{n\times d}$ such that $\Psi=\widetilde{\Psi}\sigma$ holds on $[T,T+\theta]$.

(ii) For any $(t,x,x',y,y',z,z',\mu,\mu')\in[0,T]\times(\mathbb{R}^d)^2\times(\mathbb{R}^n)^2\times(\mathbb{R}^{n\times q})^2\times\mathcal{P}_2(E)^2$, we have

$$\big|f(t,x,y,z,\mu)-f(t,x',y',z',\mu')\big|\le\Lambda(|x-x'|+|y-y'|+|z-z'|+W_2(\mu,\mu')).$$

In addition, $f(\cdot,0,0,0,m_\cdot(0,0))\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n)$.

(iii) For every $t\in[0,T+\theta]$,$\mathcal{H}^2(\mathfrak{F}^t,\mathbb{R}^n\times\mathbb{R}^{n\times q})$ into $\mathcal{H}^0(\mathfrak{F}^t,\mathcal{P}_2(E))$, and Assumption 2.1(iii) holds with $\mathfrak{F}$ replaced by $\mathfrak{F}^t$, with the same kernel ρ and constant Λ.

(iv) The maps b and σ are continuous on $[0,T+\theta]\times\mathbb{R}^d$ and, for any $(t,x,x')\in[0,T+\theta]\times(\mathbb{R}^d)^2$,

$$\begin{aligned}&\big|b(t,x)-b(t,x')\big|+\big|\sigma(t,x)-\sigma(t,x')\big|\le\Lambda|x-x'|,\\&|b(t,x)|+|\sigma(t,x)|\le\Lambda(1+|x|).\end{aligned}$$

We denote by $\mathfrak{F}^t=(\mathfrak{F}^t_s)_{0\le s\le T+\theta}$ the filtration such that, for all $0\le s\le T+\theta$, $\mathfrak{F}^t_s$ is the usual augmentation of $\sigma(W_r-W_t,t\le r\le t\vee s)$. Under Assumption 3.1(iv), Touzi (2013, Theorems 2.2 and 2.4) show that $X^{t,x}$ is uniquely defined in $\mathcal{S}^2(\mathfrak{F}^t,\mathbb{R}^d)\subset\mathcal{H}^2(\mathfrak{F}^t,\mathbb{R}^d)$, satisfying

$$\|X^{t,x}\|^2_{\mathcal{S}^2}\le C(1+|x|^2), \tag{23}$$

$$\|X^{t,x}-X^{t',x'}\|^2_{\mathcal{S}^2}\le C(|x-x'|^2+(1+|x|^2+|x'|^2)|t-t'|). \tag{24}$$

3.1 Existence, Uniqueness and Markovian estimates

In the next result, we show how Assumption 3.1 implies Assumption 2.1, so that Theorem 2.7 applies to the Markovian case.

Theorem 3.1. *Under Assumption 3.1, for any $(t,x)\in[0,T]\times\mathbb{R}^d$, the system (21) admits a unique $\mathcal{S}^2(\mathfrak{F}^t,\mathbb{R}^d)\times\mathcal{S}^2(\mathfrak{F}^t,\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F}^t,\mathbb{R}^{n\times q})$ solution.*

Proof. Let $\Phi^{t,x}=\Phi(\cdot,X^{t,x}_\cdot)$, $\Psi^{t,x}=\Psi(\cdot,X^{t,x}_\cdot)$ and, for $(\omega,s,y,z,\mu)\in\Omega\times[0,T]\times\mathbb{R}^n\times\mathbb{R}^{n\times q}\times\mathcal{P}_2(E)$, $f^{t,x}(\omega,s,y,z,\mu)=f(s,X^{t,x}_s(\omega),y,z,\mu)$. Note that $f^{t,x}$ is $Prog^t\otimes\mathcal{B}^n\otimes\mathcal{B}^{n\times q}\otimes\mathcal{B}(\mathcal{P}_2(E))$ measurable, where $Prog^t$ is the progressive sigma-algebra associated with $\mathfrak{F}^t$. Then the equations (21) for $(Y^{t,x},Z^{t,x})$ are of the form (2) with initial time 0 replaced by t, filtration $\mathfrak{F}$ replaced by $\mathfrak{F}^t$, Brownian motion W replaced by $(W_s-W_t)_{t\le s\le T+\theta}$, and parameters $(f^{t,x},\Phi^{t,x},\Psi^{t,x},m)$. Note that by Assumption 3.1(iii), the restriction of m to $[t,T+\theta]$ is independent of the values of the processes before t. The proof of Theorem 2.7 will then apply verbatim, with integrals over $[t,T]$ instead of $[0,T]$.

We now need to prove that Assumption 2.1 holds (with filtration $\mathfrak{F}^t$) in order to apply Theorem 2.7.

Under Assumption 3.1(i), $\Phi^{t,x}$ and $\Psi^{t,x}$ are $\mathfrak{F}^t$ adapted and continuous, and (23) yields

$$\begin{aligned}\|\Phi^{t,x}\|^2_{\mathcal{S}^2(T,T+\theta)}&=\mathbb{E}\left[\sup_{T\le s\le T+\theta}|\Phi(s,X^{t,x}_s)|^2\right]\\&\le 2\mathbb{E}\left[\sup_{T\le s\le T+\theta}|\Phi(s,X^{t,x}_s)-\Phi(s,0)|^2\right]+2\mathbb{E}\left[\sup_{T\le s\le T+\theta}|\Phi(s,0)|^2\right]\\&\le 2\Lambda^2\|X^{t,x}\|^2_{\mathcal{S}^2}+2\|\Phi(\cdot,0)\|^2_{\mathcal{S}^2(T,T+\theta)}<\infty,\end{aligned}$$

as well as

$$\begin{aligned}\|\Psi^{t,x}\|^2_{\mathcal{H}^2(T,T+\theta)} &= \mathbb{E}\left[\int_T^{T+\theta} |\Psi(s,X_s^{t,x})|^2 ds\right] \\ &\le 2\mathbb{E}\left[\int_T^{T+\theta} |\Psi(s,X_s^{t,x}) - \Psi(s,0)|^2 ds\right] + 2\mathbb{E}\left[\int_T^{T+\theta} |\Psi(s,0)|^2 ds\right] \\ &\le 2\Lambda^2 \|X^{t,x}\|^2_{\mathcal{H}^2(T,T+\theta)} + 2\|\Psi(\cdot,0)\|^2_{\mathcal{H}^2(T,T+\theta)} < \infty.\end{aligned}$$

Hence Assumption 2.1(i) holds.

We have, for $(\omega, s, y, y', z, z', \mu, \mu') \in \Omega \times [0,T] \times (\mathbb{R}^n)^2 \times (\mathbb{R}^{n\times q})^2 \times (\mathcal{P}_2(E))^2$,

$$\begin{aligned}\left|f^{t,x}(\omega,s,y,z,\mu) - f^{t,x}(\omega,s,y',z',\mu')\right| &= \left|f(s,X_s^{t,x}(\omega),y,z,\mu) - f(s,X_s^{t,x}(\omega),y',z',\mu')\right| \\ &\le \Lambda\left(|y-y'| + |z-z'| + W_2(\mu,\mu')\right), \\ \mathbb{E}\left[\int_t^T |f^{t,x}(s,0,0,m_s(0,0))|^2 ds\right] &= \mathbb{E}\left[\int_t^T |f(s,X_s^{t,x},0,0,m_s(0,0))|^2 ds\right] \\ &\le 2\Lambda^2\|X^{t,x}\|^2_{\mathcal{H}^2(t,T)} + 2\|f(\cdot,0,0,0,m_\cdot(0,0))\|^2_{\mathcal{H}^2(t,T)} < \infty,\end{aligned}$$

so that Assumption 2.1(ii) holds, while Assumption 2.1(iii) holds by Assumption 3.1(iii).
In conclusion, $(f^{t,x}, \Phi^{t,x}, \Psi^{t,x}, m)$ satisfies Assumption 2.1. Theorem 2.7 thus ensures existence and uniqueness of an $\mathcal{S}^2(\mathfrak{F}^t, \mathbb{R}^n) \times \mathcal{H}^2(\mathfrak{F}^t, \mathbb{R}^{n\times q})$ solution $(Y^{t,x}, Z^{t,x})$ to the Markovian ABSDE (21). The solution on $[t, T+\theta]$ is then uniquely extended to $[0, T+\theta]$ as explained below (21). □

Using the a priori estimates (12)-(13), we also obtain the following quadratic growth together with Hölder in time and Lipschitz in space dependence with respect to the initial condition.

Proposition 3.2. *Under Assumption 3.1, for $(t,x) \in [0,T] \times \mathbb{R}^d$, let $(X^{t,x}, Y^{t,x}, Z^{t,x})$ be the unique $\mathcal{S}^2(\mathfrak{F}^t, \mathbb{R}^d) \times \mathcal{S}^2(\mathfrak{F}^t, \mathbb{R}^n) \times \mathcal{H}^2(\mathfrak{F}^t, \mathbb{R}^{n\times q})$ solution to (21) as per Theorem 3.1. Then there exists a constant $C \ge 0$ such that, for all $(t,x) \in [0,T] \times \mathbb{R}^d$,*

$$\|Y^{t,x}\|^2_{\mathcal{S}^2} + \|Z^{t,x}\|^2_{\mathcal{H}^2} \le C(1+|x|^2). \tag{25}$$

Moreover, there exists a constant $C' \ge 0$ such that, for all $(t,x),(t',x') \in [0,T] \times \mathbb{R}^d$,

$$\|Y^{t,x} - Y^{t',x'}\|^2_{\mathcal{S}^2} + \|Z^{t,x} - Z^{t',x'}\|^2_{\mathcal{H}^2} \le C'(|x-x'|^2 + (1+|x|^2+|x'|^2)|t-t'|). \tag{26}$$

Proof. Note that $(Y,Z) = (0,0)$ is the unique $\mathcal{S}^2(\mathfrak{F}^t, \mathbb{R}^n) \times \mathcal{H}^2(\mathfrak{F}^t, \mathbb{R}^{n\times q})$ solution to (2) with parameters $(f, \Phi, \Psi, m) = (0,0,0,m)$.
By construction of the deterministic extension on $[0,t)$, the extended pair $(Y^{t,x}, Z^{t,x})$ solves (2) on $[0, T+\theta]$, with parameters $(f^{t,x}, \Phi^{t,x}, \Psi^{t,x}, m)$.
From the a priori estimates (12)-(13), with β large enough, we obtain the existence of $K \ge 0$ such that

$$\begin{aligned}&\|Y^{t,x}\|^2_{\mathcal{S}^2} + \|Z^{t,x}\|^2_{\mathcal{H}^2} \le \|Y^{t,x}\|^2_{\mathcal{S}^2(T)} + \|Y^{t,x}\|^2_{\mathcal{S}^2(T,T+\theta)} + \|Z^{t,x}\|^2_{\mathcal{H}^2_\beta(T)} + \|Z^{t,x}\|^2_{\mathcal{H}^2(T,T+\theta)} \\ &= \|Y^{t,x}\|^2_{\mathcal{S}^2(T)} + \|\Phi(\cdot,X_\cdot^{t,x})\|^2_{\mathcal{S}^2(T,T+\theta)} + \|Z^{t,x}\|^2_{\mathcal{H}^2_\beta(T)} + \|\Psi(\cdot,X_\cdot^{t,x})\|^2_{\mathcal{H}^2(T,T+\theta)} \\ &\le C\left(\|f(\cdot,X_\cdot^{t,x},0,0,m_\cdot(0,0))\|^2_{\mathcal{H}^2(T)} + \|\Phi(\cdot,X_\cdot^{t,x})\|^2_{\mathcal{S}^2(T,T+\theta)} + \|\Psi(\cdot,X_\cdot^{t,x})\|^2_{\mathcal{H}^2(T,T+\theta)}\right).\end{aligned}$$

We then compute, using Assumption 3.1,

$$\begin{aligned}
&\|Y^{t,x}\|^2_{\mathcal{S}^2} + \|Z^{t,x}\|^2_{\mathcal{H}^2} \\
&\leq 2C\left(\Lambda^2\|X^{t,x}\|^2_{\mathcal{H}^2(T)} + \|f(\cdot,0,0,0,m_\cdot(0,0))\|^2_{\mathcal{H}^2(T)} + \Lambda^2\|X^{t,x}\|^2_{\mathcal{S}^2(T,T+\theta)}\right. \\
&\qquad \left. +\|\Phi(\cdot,0)\|^2_{\mathcal{S}^2(T,T+\theta)} + \Lambda^2\|X^{t,x}\|^2_{\mathcal{H}^2(T,T+\theta)} + \|\Psi(\cdot,0)\|^2_{\mathcal{H}^2(T,T+\theta)}\right) \\
&\leq 2C\left(\Lambda^2(T+\theta+1)\|X^{t,x}\|^2_{\mathcal{S}^2} + \|f(\cdot,0,0,0,m_\cdot(0,0))\|^2_{\mathcal{H}^2(T)}\right. \\
&\qquad \left. +\|\Phi(\cdot,0)\|^2_{\mathcal{S}^2(T,T+\theta)} + \|\Psi(\cdot,0)\|^2_{\mathcal{H}^2(T,T+\theta)}\right) \\
&\leq C(1+|x|^2),
\end{aligned}$$

where we used (23) for the last inequality. This proves (25).

To derive (26), we compute, similarly to (12)-(13):

$$\begin{aligned}
&\|Y^{t,x} - Y^{t',x'}\|^2_{\mathcal{S}^2} + \|Z^{t,x} - Z^{t',x'}\|^2_{\mathcal{H}^2} \\
\leq&\|Y^{t,x} - Y^{t',x'}\|^2_{\mathcal{S}^2(T)} + \|\Phi(\cdot,X^{t,x}) - \Phi(\cdot,X^{t',x'})\|^2_{\mathcal{S}^2(T,T+\theta)} \\
&+ \|Z^{t,x} - Z^{t',x'}\|^2_{\mathcal{H}^2(T)} + \|\Psi(\cdot,X^{t,x}) - \Psi(\cdot,X^{t',x'})\|^2_{\mathcal{H}^2(T,T+\theta)} \\
\leq&C\left(\|f(\cdot,X^{t,x},Y^{t',x'},Z^{t',x'},m_\cdot(Y^{t',x'},Z^{t',x'})) - f(\cdot,X^{t',x'},Y^{t',x'},Z^{t',x'},m_\cdot(Y^{t',x'},Z^{t',x'}))\|^2_{\mathcal{H}^2(T)}\right. \\
&\left.+\|\Phi(\cdot,X^{t,x}) - \Phi(\cdot,X^{t',x'})\|^2_{\mathcal{S}^2(T,T+\theta)} + \|\Psi(\cdot,X^{t,x}) - \Psi(\cdot,X^{t',x'})\|^2_{\mathcal{H}^2(T,T+\theta)}\right) \\
&\leq C\Lambda^2\left(\|X^{t,x} - X^{t',x'}\|^2_{\mathcal{H}^2(T)} + \|X^{t,x} - X^{t',x'}\|^2_{\mathcal{S}^2(T,T+\theta)} + \|X^{t,x} - X^{t',x'}\|^2_{\mathcal{H}^2(T,T+\theta)}\right) \\
&\leq C\Lambda^2(T+\theta+1)\|X^{t,x} - X^{t',x'}\|^2_{\mathcal{S}^2} \leq C'(|x-x'|^2 + (1+|x|^2+|x'|^2)|t-t'|),
\end{aligned}$$

where we used (24) for the last inequality. □

3.2 Semigroup Properties

In this section, we prove that, in this Markovian setting, for each $0 \leq t \leq s \leq T+\theta$ and $x \in \mathbb{R}^d$, $(Y^{t,x}_s, Z^{t,x}_s)$ can be represented as a measurable function of time s and state $X^{t,x}_s$. This is similar to what is obtained for the classical BSDEs without anticipated feature in El Karoui et al. (1997), modulo the following result, which proves that the anticipated term is itself nothing but a ($\mathcal{P}_2(E)$ valued) function of time and state.

As in El Karoui et al. (1997) following Çinlar et al. (1980, Convention (3.6)-(ii) and Theorem (6.27)), let $\mathcal{E}$ denote the sigma algebra on $\mathbb{R}^d$ generated by the α excessive functions relative to X (see Blumenthal and Getoor (2007, Chapter II)).

In order to deal with the anticipated term, we need the following additional assumption, which we prove to be satisfied in a fairly general setting.

Assumption 3.2. For any $\mathcal{B}([0,T+\theta])\otimes\mathcal{E}$ measurable $(u,v) : [0,T+\theta]\times\mathbb{R}^d \to \mathbb{R}^n \times \mathbb{R}^{n\times q}$ such that, for any $(t_0,x_0) \in [0,T]\times\mathbb{R}^d$, $(u(\cdot,X^{t_0,x_0}_\cdot), v(\cdot,X^{t_0,x_0}_\cdot)) \in \mathcal{H}^2(\mathfrak{F}^{t_0}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$, there exists a $\mathcal{B}([0,T])\otimes\mathcal{E}$ measurable map $m^{u,v} : [0,T]\times\mathbb{R}^d \to \mathcal{P}_2(E)$ such that, for any $(t_0,x_0) \in [0,T]\times\mathbb{R}^d$,

$$m_\cdot(u(\cdot,X^{t_0,x_0}_\cdot), v(\cdot,X^{t_0,x_0}_\cdot)) = m^{u,v}(\cdot,X^{t_0,x_0}_\cdot), \quad \text{a.e. on } [t_0,T]\times\Omega.$$

Lemma 3.3. *Assume that, for all (u,v) as in Assumption 3.2, there exists a measurable map $A^{u,v} : [0,T]\times\mathbb{R}^d\times\mathcal{C}([0,T+\theta],\mathbb{R}^q) \to E$ such that for all $(t_0,x_0) \in [0,T]\times\mathbb{R}^d$, $\mathcal{L}(A^{u,v}(t_0,x_0,(W-W_{t_0})\mathbf{1}_{[t_0,T+\theta]})) \in \mathcal{P}_2(E)$ and $m_t(u(\cdot,X^{t_0,x_0}_\cdot), v(\cdot,X^{t_0,x_0}_\cdot))(\omega) = \mathcal{L}(A^{u,v}(t,X^{t_0,x_0}_t,(W-W_t)\mathbf{1}_{[t,T+\theta]})|\mathfrak{F}_t)(\omega)$ for almost every $(t,\omega) \in [t_0,T]\times\Omega$.*
Then m satisfies Assumption 3.2, with $m^{u,v}$ defined by

$$m^{u,v}(t,x) = \mathcal{L}(A^{u,v}(t,x,(W-W_t)\mathbf{1}_{[t,T+\theta]})),\ (t,x) \in [0,T]\times\mathbb{R}^d. \tag{27}$$

Proof. Fix a map (u,v) as in Assumption 3.2 and consider $(t_0,x_0)\in[0,T]\times\mathbb{R}^d$, $t\in[t_0,T]$ and $m^{u,v}$ as in (27).
Since $(t,\omega)\mapsto(W-W_t)(\omega)\mathbf{1}_{[t,T+\theta]}$ is measurable, the map $(t,x)\mapsto m^{u,v}(t,x)$ is measurable. For any bounded continuous map $\varphi:E\to\mathbb{R}$, we have, almost everywhere on $[t_0,T]\times\Omega$,

$$\begin{aligned}\int_E\varphi(e)m_t(u(\cdot,X_\cdot^{t_0,x_0}),v(\cdot,X_\cdot^{t_0,x_0}))(de)&=\mathbb{E}\left[\varphi(A^{u,v}(t,X_t^{t_0,x_0},(W-W_t)\mathbf{1}_{[t,T+\theta]}))|\mathfrak{F}_t\right]\\&=\mathbb{E}\left[\varphi(A^{u,v}(t,x,(W-W_t)\mathbf{1}_{[t,T+\theta]}))\right]_{\big|x=X_t^{t_0,x_0}}\\&=\int_E\varphi(e)m^{u,v}(t,X_t^{t_0,x_0})(de),\end{aligned}$$

where we used the fact that $X_t^{t_0,x_0}$ is $\mathfrak{F}_t$ measurable while $(W-W_t)\mathbf{1}_{[t,T+\theta]}$ is independent from $\mathfrak{F}_t$ for the second equality, and the definition of $m^{u,v}$ for the last equality. This proves (considering a countable family of test functions that separate measures as in Lemma 2.1) that $m_t(u(\cdot,X_\cdot^{t_0,x_0}),v(\cdot,X_\cdot^{t_0,x_0}))(\omega)=m^{u,v}(t,X_t^{t_0,x_0}(\omega))$ a.e., as desired. □

Proposition 3.4. *Assume that, for every $(Y,Z)\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})$, $m_t(Y,Z)(\omega)=\mathcal{L}(A(t,(Y,Z,W-W_t)\mathbf{1}_{[t,T+\theta]})|\mathfrak{F}_t)(\omega)$ almost everywhere, where $A:[0,T]\times L^2([0,T+\theta],\mathbb{R}^n)\times L^2([0,T+\theta],\mathbb{R}^{n\times q})\times\mathcal{C}([0,T+\theta],\mathbb{R}^q)\to E$ is a measurable map such that, for all (u,v) as in Assumption 3.2 and $(t,x)\in[0,T]\times\mathbb{R}^d$, $\mathcal{L}(A(t,(u(\cdot,X_\cdot^{t,x}),v(\cdot,X_\cdot^{t,x}),W_\cdot-W_t)\mathbf{1}_{[t,T+\theta]}))\in\mathcal{P}_2(E)$. Then m satisfies Assumption 3.2, with, for any measurable (u,v) as in Assumption 3.2, $m^{u,v}$ defined by*

$$m^{u,v}(t,x)=\mathcal{L}(A(t,(u(\cdot,X_\cdot^{t,x}),v(\cdot,X_\cdot^{t,x}),W_\cdot-W_t)\mathbf{1}_{[t,T+\theta]})),\ (t,x)\in[0,T]\times\mathbb{R}^d.$$

Proof. First, X being a strong solution to the first equation in (21), by Touzi (2013, Section 2.2, p.10), there exists a measurable $\psi:[0,T]\times\mathbb{R}^d\times\mathcal{C}([0,T+\theta],\mathbb{R}^q)\to\mathcal{C}([0,T+\theta],\mathbb{R}^d)$ such that, for any $(t,x)\in[0,T]\times\mathbb{R}^d$,

$$X_\cdot^{t,x}=\psi(t,x,W_{\cdot\vee t}-W_t).\tag{28}$$

By the flow property, $X^{t_0,x_0}\mathbf{1}_{[t,T+\theta]}$ is indistinguishable from $X^{t,X_t^{t_0,x_0}}\mathbf{1}_{[t,T+\theta]}$, hence, for any (u,v) as in Assumption 3.2 and any $(t_0,x_0)\in[0,T]\times\mathbb{R}^d$, we almost everywhere have on $[t_0,T]\times\Omega$,

$$\begin{aligned}&m_t(u(\cdot,X_\cdot^{t_0,x_0}),v(\cdot,X_\cdot^{t_0,x_0}))\\&=\mathcal{L}(A(t,(u(\cdot,X_\cdot^{t_0,x_0}),v(\cdot,X_\cdot^{t_0,x_0}),W_\cdot-W_t)\mathbf{1}_{[t,T+\theta]})|\mathfrak{F}_t)\\&=\mathcal{L}(A(t,(u(\cdot,X_\cdot^{t,X_t^{t_0,x_0}})),v(\cdot,X_\cdot^{t,X_t^{t_0,x_0}}),W_\cdot-W_t)\mathbf{1}_{[t,T+\theta]})|\mathfrak{F}_t)\\&=\mathcal{L}(A(t,(u(\cdot,\psi(t,X_t^{t_0,x_0},W_{\cdot\vee t}-W_t)),v(\cdot,\psi(t,X_t^{t_0,x_0},W_{\cdot\vee t}-W_t)),W_\cdot-W_t)\mathbf{1}_{[t,T+\theta]})|\mathfrak{F}_t)\\&=\mathcal{L}(A^{u,v}(t,X_t^{t_0,x_0},(W_\cdot-W_t)\mathbf{1}_{[t,T+\theta]})|\mathfrak{F}_t),\end{aligned}$$

with $A^{u,v}(t,x,w):=A(t,(u(\cdot,\psi(t,x,w)),v(\cdot,\psi(t,x,w)),w)\mathbf{1}_{[t,T+\theta]})$, which is well defined for $\mathcal{L}((W-W_t)\mathbf{1}_{[t,T+\theta]})$ almost every w, and we extend it measurably and arbitrarily to the whole path space. The result then follows from Lemma 3.3. □

Proposition 3.5. *Assume that $E=\mathbb{R}^m$ and that, for every $(Y,Z)\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})$, $m_t(Y,Z)(\omega)=\mathcal{L}(\int_t^{t+\theta}A(s,Y_s,Z_s)dW_s|\mathfrak{F}_t)(\omega)$ for almost every $(t,\omega)\in[0,T]\times\Omega$, where $A:[0,T+\theta]\times\mathbb{R}^n\times\mathbb{R}^{n\times q}\to\mathbb{R}^{m\times q}$ is a measurable map such that $A(\cdot,Y,Z)\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^{m\times q})$ for all $(Y,Z)\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})$, where $m\geq1$ is an integer. Then m satisfies Assumption 3.2, with, for any (u,v) as in Assumption 3.2 , $m^{u,v}$ defined by*

$$m^{u,v}(t,x)=\mathcal{L}\left(\int_t^{t+\theta}A(s,u(s,X_s^{t,x}),v(s,X_s^{t,x}))dW_s\right),\ (t,x)\in[0,T]\times\mathbb{R}^d.$$

Proof. For any (u,v) as in Assumption 3.2 and any $(t_0,x_0)\in[0,T]\times\mathbb{R}^d$, we almost everywhere have, on $[t_0,T]\times\Omega$, using the same arguments and notations as in the previous proof,

$$\begin{aligned}
&m_t(u(\cdot,X_\cdot^{t_0,x_0}),v(\cdot,X_\cdot^{t_0,x_0}))\\
&=\mathcal{L}\left(\int_t^{t+\theta}A(s,u(s,X_s^{t_0,x_0}),v(s,X_s^{t_0,x_0}))dW_s|\mathfrak{F}_t\right)\\
&=\mathcal{L}\left(\int_t^{t+\theta}A(s,u(s,\psi(t,X_t^{t_0,x_0},W_{\cdot\vee t}-W_t)_s),v(s,\psi(t,X_t^{t_0,x_0},W_{\cdot\vee t}-W_t)_s))dW_s|\mathfrak{F}_t\right).
\end{aligned}$$

Observe that the inner random variable is $\sigma(X_t^{t_0,x_0},(W_s-W_t)_{s\in[t,t+\theta]})$ measurable, hence of the form $A^{u,v}(t,X_t^{t_0,x_0},(W_\cdot-W_t)\mathbf{1}_{[t,T+\theta]})$ for some measurable map $A^{u,v}:[0,T]\times\mathbb{R}^d\times\mathcal{C}([0,T+\theta],\mathbb{R}^q)\to\mathbb{R}^m$. By Itô's isometry and the assumption on A, for every (t,x) the random variable $\int_t^{t+\theta}A(s,u(s,X_s^{t,x}),v(s,X_s^{t,x}))\,dW_s$ belongs to $L^2(\Omega,\mathbb{R}^m)$. The result then follows from Lemma 3.3. □

We then have the following representation result.

Theorem 3.6. *Under Assumptions 3.1 and 3.2, there exists a $\mathcal{B}([0,T+\theta])\otimes\mathcal{E}$ measurable map*

$$(u,v):[0,T+\theta]\times\mathbb{R}^d\to\mathbb{R}^n\times\mathbb{R}^{n\times d}$$

such that, for any $(t,x)\in[0,T]\times\mathbb{R}^d$, $u(t,x)=Y_t^{t,x}$ and, $ds\otimes d\mathbb{P}$ almost surely on $[t,T+\theta]$,

$$Y_s^{t,x}=u(s,X_s^{t,x})\quad\text{ and }\quad Z_s^{t,x}=v(s,X_s^{t,x})\sigma(s,X_s^{t,x}).\tag{29}$$

Moreover, the function u is globally Lipschitz continuous in its space variables and locally $\frac{1}{2}$-Hölder continuous in its time variable on $[0,T]\times\mathbb{R}^d$.

Proof. For any $(t,x)\in[0,T]\times\mathbb{R}^d$ and $(\widetilde{Y},\widetilde{Z})\in\mathcal{H}^2(\mathfrak{F}^t,\mathbb{R}^n\times\mathbb{R}^{n\times q})$, we define $\Gamma^{t,x}(\widetilde{Y},\widetilde{Z})=(Y,Z)$ as follows: $(Y,Z)_{|[t,T]}$ is defined as the unique $\mathcal{H}^2((\mathfrak{F}^t)^T,\mathbb{R}^n\times\mathbb{R}^{n\times q})$ solution to the classical BSDE

$$Y_s=\Phi(T,X_T^{t,x})+\int_s^T f(r,X_r^{t,x},Y_r,Z_r,m_r(\widetilde{Y},\widetilde{Z}))dr-\int_s^T Z_r dW_r,\quad s\in[t,T],$$

we set $Z=0$ on $[0,t)$ and $Y_{|[0,t]}$ is the solution to the ODE

$$Y_s=Y_t+\int_s^t f(r,x,Y_r,0,m_r(\widetilde{Y},\widetilde{Z}))dr,\quad s\in[0,t],$$

and we eventually set $(Y,Z)=(\Phi(\cdot,X_\cdot^{t,x}),\Psi(\cdot,X_\cdot^{t,x}))$ on $[T,T+\theta]$.

From Theorem 2.7(ii), we know that the sequence of processes $(Y^{t,x,k},Z^{t,x,k})_{k\geq0}$ defined by $(Y^{t,x,0},Z^{t,x,0})=(0,0)$ and, for $k\geq0$, $(Y^{t,x,k+1},Z^{t,x,k+1})=\Gamma^{t,x}(Y^{t,x,k},Z^{t,x,k})$, converges almost everywhere to $(Y^{t,x},Z^{t,x})$ on $[t,T]$, hence on $[t,T+\theta]$ (as all iterates with $k\geq1$ coincide on $[T,T+\theta]$).

We now prove by induction that, for any $k\geq0$, $Y_s^{t,x,k}=u_k(s,X_s^{t,x})$ and $Z_s^{t,x,k}=(v_k\sigma)(s,X_s^{t,x})$ almost everywhere on $[t,T+\theta]$ for some $\mathcal{B}([0,T+\theta])\otimes\mathcal{E}$ measurable map $(u_k,v_k):[0,T+\theta]\times\mathbb{R}^d\to\mathbb{R}^n\times\mathbb{R}^{n\times d}$.

Since $(Y^{t,x,0},Z^{t,x,0})=(0,0)$, the property is satisfied for $k=0$. Assume that the property holds for some $k\geq0$. From Assumption 3.2 applied to u_k and $v_k\sigma$, we get $m_s(Y^{t,x,k},Z^{t,x,k})=m^k(s,X_s^{t,x})=m^{u_k,v_k\sigma}(s,X_s^{t,x})$ where m^k is a $\mathcal{P}_2(E)$ valued measurable map. Then, on $[t,T]$, $(Y^{t,x,k+1},Z^{t,x,k+1})$ is the unique $\mathcal{S}^2(\mathfrak{F}^t,\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F}^t,\mathbb{R}^{n\times q})$ solution to the classical BSDE

$$Y_s=\Phi(T,X_T^{t,x})+\int_s^T f^{k+1}(r,X_r^{t,x},Y_r,Z_r)dr-\int_s^T Z_r dW_r,$$

where

$$f^{k+1} : [0,T] \times \mathbb{R}^d \times \mathbb{R}^n \times \mathbb{R}^{n\times q} \to \mathbb{R}^n$$
$$(t,x,y,z) \mapsto f(t,x,y,z,m^k(t,x)).$$

Note that for every $(t,x) \in [0,T]\times\mathbb{R}^d$, $f^{k+1}(\cdot, X^{t,x}_\cdot, 0, 0) \in \mathcal{H}^2((\mathfrak{F}^t)^T, \mathbb{R}^n)$ by Assumption 2.1 and the admissibility of the Picard iterates. Applying El Karoui et al. (1997, Theorem 4.1), we obtain the existence of a $\mathcal{B}([0,T])\otimes\mathcal{E}$ measurable map $(u_{k+1}, v_{k+1}) : [0,T]\times\mathbb{R}^d \to \mathbb{R}^n \times \mathbb{R}^{n\times d}$ such that, for all $(t,x) \in [0,T]\times\mathbb{R}^d$, $Y_t^{t,x,k+1} = u_{k+1}(t,x)$ and $(Y_s^{t,x,k+1}, Z_s^{t,x,k+1}) = (u_{k+1}, v_{k+1}\sigma)(s, X_s^{t,x})$, almost everywhere on $[t,T]$. We eventually set $(u_{k+1}, v_{k+1})(s,x) = (\Phi(s,x), \widetilde{\Psi}(s,x))$ for $(s,x) \in [T, T+\theta]\times\mathbb{R}^d$ (recall Assumption 3.1(i)), so $(Y^{t,x,k+1}, Z^{t,x,k+1}) = (u_{k+1}, v_{k+1}\sigma)(\cdot, X^{t,x})$ holds almost everywhere on $[t, T+\theta]$.

We define maps $u : [0,T+\theta]\times\mathbb{R}^d \to \mathbb{R}^n$ and $z : [0,T+\theta]\times\mathbb{R}^d \to \mathbb{R}^{n\times q}$ componentwise by

$$u^i(t,x) = \begin{cases} \lim_{k\to\infty} u_k^i(t,x) & \text{if it exists in } \mathbb{R} \\ 0 & \text{otherwise} \end{cases},$$
$$z^{i,j}(t,x) = \begin{cases} \lim_{k\to\infty} (v_k^\top \sigma)^{i,j}(t,x) & \text{if it exists in } \mathbb{R} \\ 0 & \text{otherwise.} \end{cases}$$

Since u_k and $v_k\sigma$ are $\mathcal{B}([0,T+\theta])\otimes\mathcal{E}$ measurable, so are u and z. Moreover, by the $ds\otimes d\mathbb{P}$ a.e. convergence of $(Y^{t,x,k}, Z^{t,x,k})$ towards $(Y^{t,x}, Z^{t,x})$, we have, for any $(t,x) \in [0,T]\times\mathbb{R}^d$, almost everywhere on $[t, T+\theta]$,

$$Y_s^{t,x,i} = \lim_{k\to\infty} Y_s^{t,x,k,i} = \lim_{k\to\infty} u_k^i(s, X_s^{t,x}) = u^i(s, X_s^{t,x}),$$
$$Z_s^{t,x,i,j} = \lim_{k\to\infty} Z_s^{t,x,k,i,j} = \lim_{k\to\infty} (v_k\sigma)^{i,j}(s, X_s^{t,x}) = z^{i,j}(s, X_s^{t,x}).$$

Hence we have, almost everywhere on $[t, T+\theta]$,

$$u(s, X_s^{t,x}) = Y_s^{t,x} \quad \text{and} \quad z(s, X_s^{t,x}) = Z_s^{t,x}.$$

Now set $v = z\sigma^\dagger$, where $\sigma^\dagger$ denotes the Moore-Penrose pseudo-inverse of σ; this map is $\mathcal{B}([0,T+\theta])\otimes\mathcal{E}$ measurable. Since, for every k,

$$(v_k\sigma)(s, X_s^{t,x}) = (v_k\sigma)(s, X_s^{t,x})\, \sigma(s, X_s^{t,x})^\dagger \sigma(s, X_s^{t,x}),$$

passing to the limit yields, $ds\otimes d\mathbb{P}$ almost surely on $[t, T+\theta]$,

$$z(s, X_s^{t,x}) = (z\sigma^\dagger)(s, X_s^{t,x})\sigma(s, X_s^{t,x}) = (v\sigma)(s, X_s^{t,x}).$$

Finally, by Theorem 2.7(ii), since $Y_t^{t,x,k}$ and $Y_t^{t,x}$ are $\mathfrak{F}_t^t$ measurable, hence deterministic, we have

$$|u_k(t,x) - Y_t^{t,x}|^2 = |Y_t^{t,x,k} - Y_t^{t,x}|^2 \le \|Y^{t,x,k} - Y^{t,x}\|^2_{\mathcal{S}^2(t,T)} \to 0.$$

Therefore $u(t,x) = \lim_{k\to\infty} u_k(t,x) = Y_t^{t,x}$.

Finally, let $(t,t',x,x') \in [0,T]^2 \times (\mathbb{R}^d)^2$, and assume e.g. $t \le t'$. By Proposition 3.2, there exists $C \ge 0$ such that

$$|u(t,x) - u(t,x')|^2 = |Y_t^{t,x} - Y_t^{t,x'}|^2 \le \|Y^{t,x} - Y^{t,x'}\|^2_{\mathcal{S}^2}$$
$$\le C|x - x'|^2.$$

Thus $u(t,\cdot)$ is globally Lipschitz continuous, uniformly with respect to t. Moreover, one has, using

$$\begin{aligned}
|u(t,x)-u(t',x)|^2 &= |Y_t^{t,x}-Y_{t'}^{t',x}|^2 \\
&\le 2|Y_t^{t,x}-Y_t^{t',x}|^2+2|Y_t^{t',x}-Y_{t'}^{t',x}|^2 \\
&\le 2\|Y^{t,x}-Y^{t',x}\|^2_{\mathcal{S}^2(t,T)}+2|Y_t^{t',x}-Y_{t'}^{t',x}|^2 \\
&\le 2C(1+|x|^2)|t-t'|+2|t-t'|\mathbb{E}\left[\int_t^{t'}|f(r,x,Y_r^{t',x},0,m_r(Y^{t',x},Z^{t',x}))|^2dr\right] \\
&\le C(1+|x|^2)|t-t'|,
\end{aligned}$$

where we used Proposition 3.2 for the first term, Cauchy-Schwarz inequality and the a priori estimates for the second term. This proves that $u(\cdot,x)$ is locally $\frac{1}{2}$-Hölder continuous. □

3.3 Associated nonlocal PDE and verification

We denote by $\mathcal{A}$ the infinitesimal generator of X, namely, for smooth $\varphi:[0,T]\times\mathbb{R}^d\to\mathbb{R}$ with row-gradient $\partial_x\varphi$ and Hessian matrix $\partial^2_{x^2}\varphi$,

$$\mathcal{A}\varphi=\partial_x\varphi b+\frac{1}{2}\mathrm{Tr}[\sigma^\top(\partial^2_{x^2}\varphi)\,\sigma].$$

The goal of this section is to introduce a nonlocal PDE and prove, by a verification argument, that any sufficiently regular solution of this PDE provides a representation of the Markovian ABSDE solution. To this end, consider the PDE

$$\begin{aligned}
&u=\Phi \quad \text{on } [T,T+\theta]\times\mathbb{R}^d,\\
&\partial_t u+\mathcal{A}u+f(t,x,u,\partial_x u\,\sigma,m^{u,z})=0 \quad \text{on } [0,T)\times\mathbb{R}^d,
\end{aligned}\tag{30}$$

where $z=\partial_x u\,\sigma\mathbf{1}_{[0,T]}+\Psi\mathbf{1}_{(T,T+\theta]}$ and $m^{u,z}$ is defined in Assumption 3.2. Note that for vector-valued u, $\mathcal{A}$ acts component-wise.

Remark 3.1. PDEs nonlocal in space are commonly found in relation with jump-diffusions or with McKean-Vlasov or mean-field control PDEs (Barles et al., 1997; Pham and Wei, 2017). The PDE (30), dubbed "anticipated" hereafter, is nonlocal, not only in space, but also in time, due to the anticipated nature of the BSDE: the equation valued at (t,x) involves, in general, the values of u on $[t,T+\theta]\times\mathbb{R}^d$. In general, this PDE is also implicit in the sense that, even in the specific setups of Propositions 3.4 and 3.5 covering the applications of Example 2.4, its anticipated input $u,z\mapsto m^{u,z}$ is only characterized in these propositions mathematically, but it is not known explicitly.

The aforementioned link is provided by the following verification theorem.

Proposition 3.7 (Markovian ABSDE verification theorem)**.** *Under Assumption 3.2, suppose that the nonlocal PDE* (30) *admits a regular solution $u\in\mathcal{C}^{1,2}([0,T+\theta]\times\mathbb{R}^d;\mathbb{R}^n)$ such that (u,z) is $\mathcal{B}([0,T+\theta])\otimes\mathcal{E}$ measurable, where $z:=\partial_x u\,\sigma\mathbf{1}_{[0,T]}+\Psi\mathbf{1}_{(T,T+\theta]}$, and for any $(t,x)\in[0,T+\theta]\times\mathbb{R}^d$,*

$$|u(t,x)|+|((\partial_x u)^\top\sigma)(t,x)|\le C(1+|x|).$$

Then, under Assumption 3.1, for each $(t,x)\in[0,T]\times\mathbb{R}^d$, we have that $Y^{t,x}$ and $u(\cdot,X^{t,x}_\cdot)$ are indistinguishable on $[t,T+\theta]$, and $Z^{t,x}=(\partial_x u\,\sigma\mathbf{1}_{[0,T]}+\Psi\mathbf{1}_{(T,T+\theta]})(\cdot,X^{t,x})$ a.e. on $[t,T+\theta]\times\Omega$.

Proof. First, by the growth condition, Assumption 3.1(i), and (23), for every $(t_0,x_0)\in[0,T]\times\mathbb{R}^d$, $(u(\cdot,X^{t_0,x_0}_\cdot),z(\cdot,X^{t_0,x_0}_\cdot))$ belongs to $\mathcal{S}^2(\mathfrak{F}^{t_0},\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F}^{t_0},\mathbb{R}^{n\times q})$. Hence Assumption 3.2 applies to (u,z).

We now fix $(t,x) \in [0,T] \times \mathbb{R}^d$. Letting $(Y_s, Z_s) = (u,z)(s, X_s^{t,x})$ for $s \in [t, T+\theta]$, we first observe, by (30), that $Y_s = \Phi(s, X_s^{t,x})$ and $Z_s = \Psi(s, X_s^{t,x})$, $s \in (T, T+\theta]$.

For $s \in [t,T]$, applying Itô's formula to the smooth function u and the fact that u solves (30), we obtain

$$\begin{aligned}
&\Phi(T, X_T^{t,x}) \\
&= Y_s + \int_s^T (\partial_t + \mathcal{A})u(r, X_r^{t,x})dr + \int_s^T (\partial_x u\, \sigma)(r, X_r^{t,x})dW_r \\
&= Y_s - \int_s^T f(r, X_r^{t,x}, u(r, X_r^{t,x}), (\partial_x u\, \sigma)(r, X_r^{t,x}), m^{u,z}(r, X_r^{t,x}))dr + \int_s^T Z_r dW_r \\
&= Y_s - \int_s^T f(r, X_r^{t,x}, Y_r, Z_r, m^{u,z}(r, X_r^{t,x}))dr + \int_s^T Z_r dW_r.
\end{aligned}$$

From Assumption 3.2, we obtain,

$$m^{u,z}(r, X_r^{t,x}) = m_r(u(\cdot, X_\cdot^{t,x}), z(\cdot, X_\cdot^{t,x})) = m_r(Y, Z), \quad \text{a.e. on } [t,T] \times \Omega,$$

hence

$$\Phi(T, X_T^{t,x}) = Y_s - \int_s^T f(r, X_r^{t,x}, Y_r, Z_r, m_r(Y,Z))dr + \int_s^T Z_r dW_r.$$

This proves that (Y,Z) is a $\mathcal{S}^2(\mathfrak{F}^t, \mathbb{R}^n) \times \mathcal{H}^2(\mathfrak{F}^t, \mathbb{R}^{n\times q})$ solution to (21), hence $(Y,Z) = (Y^{t,x}, Z^{t,x})$, by Theorem 3.1. □

Remark 3.2. Conversely, in dimension onr and under a suitable comparison theorem for Markovian ABSDEs (21), one expects that a solution to (21) would provide a viscosity solution to the PDE (30). A uniqueness result for the latter would then allow characterizing the solution to a Markovian ABSDE in terms of the corresponding PDE. This is left for further research.

4 The ABSDE Implicit Time-Discretization Scheme

As an application of the above results, we study a time-discretization scheme for the Markovian ABSDE (21), in line with the one for classical BSDEs originally introduced in Bouchard and Touzi (2004). For notational simplicity, we present the time-discretization scheme in the scalar case $n = 1$, while d and q remain arbitrary. The scheme and the convergence analysis extend componentwise to arbitrary n, provided the corresponding vector-valued assumptions are imposed.

4.1 Additional Assumptions

In order to obtain a convergence result, we need to make the following additional assumptions on the data.

Assumption 4.1. Φ, Ψ, b, σ, f and m satisfy:
(i) For all $(t,s,x) \in [T, T+\theta]^2 \times \mathbb{R}^d$, $|\Phi(t,x) - \Phi(s,x)| + |\Psi(t,x) - \Psi(s,x)| \le \Lambda |t-s|^{\frac{1}{2}}$,
(ii) For all $(t,s,x) \in [0, T+\theta]^2 \times \mathbb{R}^d$, $|b(t,x) - b(s,x)| + |\sigma(t,x) - \sigma(s,x)| \le \Lambda |t-s|^{\frac{1}{2}}$,
(iii) For any $(t,s,x,y,z,\mu) \in [0,T]^2 \times \mathbb{R}^d \times \mathbb{R} \times \mathbb{R}^q \times \mathcal{P}_2(E)$,

$$|f(t,x,y,z,\mu) - f(s,x,y,z,\mu)| \le \Lambda |t-s|^{\frac{1}{2}},$$

(iv) There exists $\widetilde{m} : [0,T] \times \mathbb{R}^d \to \mathcal{P}_2(E)$ such that, for any $(t,x) \in [0,T] \times \mathbb{R}^d$, $m_s(Y^{t,x}, Z^{t,x}) = \widetilde{m}(s, X_s^{t,x})$ almost everywhere on $[t,T] \times \Omega$, and such that for any $(t,x),(s,y) \in [0,T] \times \mathbb{R}^d$,

$$W_2(\widetilde{m}(t,x), \widetilde{m}(s,y)) \le \Lambda((1 + |x| + |y|)|t-s|^{\frac{1}{2}} + |x-y|). \tag{31}$$

The first three requirements are standard assumptions regarding the coefficients. The last one involves the Markovian representation of the anticipated term (available e.g. as soon as Theorem 3.6 applies) along the ABSDE solution. We show in the lemma below that it is satisfied for (some of) the particular cases of interest to us, among those of Example 2.4.

Lemma 4.1. *Let Assumptions 3.1 and 4.1(i) hold.*
(i) *Assume* $m_t(Y,Z) = \mathcal{L}_t(Y_{t+\theta})$ *for any* $(Y,Z) \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ *and* $t \in [0,T]$. *Then Assumption 4.1(iv) holds.*
(ii) *Assume* $m_t(Y,Z) = \mathcal{L}_t(\int_t^{t+\theta} Z_s dW_s)$ *for any* $(Y,Z) \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ *and* $t \in [0,T]$. *Then Assumption 4.1(iv) holds.*

Proof. **(i)** First, by Proposition 3.4, Assumption 3.2 is satisfied. Under Assumption 3.1, Theorem 3.6 then provides a $\mathcal{B}([0,T+\theta]) \otimes \mathcal{E}$ measurable map $(u,v) : [0,T+\theta] \times \mathbb{R}^d \to \mathbb{R}^n \times \mathbb{R}^{d\times n}$ such that, for any $(t,x) \in [0,T] \times \mathbb{R}^d$, $(Y^{t,x}, Z^{t,x}) = (u, v^\top \sigma)(\cdot, X^{t,x})$. Hence, $\widetilde{m}$ of Assumption 4.1(iv) is given by $m^{u,v^\top\sigma}$ in Assumption 3.2, hence $\widetilde{m}(t,x) = \mathcal{L}(u(t+\theta, X^{t,x}_{t+\theta}))$, by Proposition 3.4. Now, let $(t,x),(s,y) \in [0,T] \times \mathbb{R}^d$. One has, using the regularity of u from Theorem 3.6 and from Assumption 4.1(i) for the regularity of Φ,

$$\begin{aligned}
W_2^2(\widetilde{m}(t,x), \widetilde{m}(s,y)) &= W_2^2(\mathcal{L}(u(t+\theta, X^{t,x}_{t+\theta})), \mathcal{L}(u(s+\theta, X^{s,y}_{s+\theta}))) \\
&\le \mathbb{E}\left[|u(t+\theta, X^{t,x}_{t+\theta}) - u(s+\theta, X^{s,y}_{s+\theta})|^2\right] \\
&\le C\left(|t-s|\mathbb{E}\left[(1+|X^{t,x}_{t+\theta}|^2) + |X^{s,y}_{s+\theta}|^2\right] + \mathbb{E}\left[|X^{t,x}_{t+\theta} - X^{s,y}_{s+\theta}|^2\right]\right) \\
&\le C((1+|x|^2+|y|^2)|t-s| + |x-y|^2),
\end{aligned}$$

which gives Assumption 4.1(iv).
(ii) From Proposition 3.5, Assumption 3.2 is satisfied. Again, Theorem 3.6 yields $(Y^{t,x}_s, Z^{t,x}_s) = (u, v^\top\sigma)(s, X^{t,x}_s)$, hence $\widetilde{m}$ of Assumption 4.1(iv) is given by $m^{u,v^\top\sigma}$ in Assumption 3.2, hence $\widetilde{m}(t,x) = \mathcal{L}(\int_t^{t+\theta} z(s, X^{t,x}_s) dW_s)$, by Proposition 3.5.

Let $(t,x),(s,y) \in [0,T] \times \mathbb{R}^d$. One has, with $I = [t, t+\theta]$ and $J = [s, s+\theta]$,

$$\begin{aligned}
W_2^2(\widetilde{m}(t,x), \widetilde{m}(s,y)) &= W_2^2(\mathcal{L}(\int_t^{t+\theta} Z^{t,x}_u dW_u), \mathcal{L}(\int_s^{s+\theta} Z^{s,y}_u dW_u)) \\
&\le \mathbb{E}\left[|\int_t^{t+\theta} Z^{t,x}_u dW_u - \int_s^{s+\theta} Z^{s,y}_u dW_u|^2\right] \\
&\le \mathbb{E}\left[\int_{I\cap J} |Z^{t,x}_u - Z^{s,y}_u|^2 du\right] + \mathbb{E}\left[\int_{I\setminus J} |Z^{t,x}_u|^2 du\right] + \mathbb{E}\left[\int_{J\setminus I} |Z^{s,y}_u|^2 du\right].
\end{aligned}$$

Regarding the first term, from (26), one gets

$$\mathbb{E}\left[\int_{I\cap J} |Z^{t,x}_u - Z^{s,y}_u|^2 du\right] \le \|Z^{t,x} - Z^{s,y}\|^2_{\mathcal{H}^2} \le C(|x-y|^2 + (1+|x|^2+|y|^2)|t-s|).$$

Regarding the two last terms, noticing that $I \setminus J$ and $J \setminus I$ are intervals of length at most $|t-s|$, estimates concluding the proof of Assumption 4.1(iv) follow from the fact that, for $t_0 \le a < b$ and $x_0 \in \mathbb{R}^d$,

$$\mathbb{E}\left[\int_a^b |Z^{t_0,x_0}_s|^2 ds\right] \le C(1+|x_0|^2)(b-a). \tag{32}$$

To prove (32), assume first that $a < b \leq T$. We compute, using Theorem 3.6, the flow property of X and the representation (28),

$$\begin{aligned}
\mathbb{E}\left[\int_a^b |Z_s^{t_0,x_0}|^2 ds\right] &= \mathbb{E}\left[\int_a^b |z(s, X_s^{t_0,x_0})|^2 ds\right] = \mathbb{E}\left[\int_a^b |z(s, X_s^{a,X_a^{t_0,x_0}})|^2 ds\right] \\
&= \mathbb{E}\left[\int_a^b |z(s, \psi_s(a, X_a^{t_0,x_0}, W_{\cdot\vee a} - W_a))|^2 ds\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\int_a^b |z(s, \psi_s(a, x, W_{\cdot\vee a} - W_a))|^2 ds\right]_{|x=X_a^{t_0,x_0}}\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\int_a^b |z(s, X_s^{a,x})|^2 ds\right]_{|x=X_a^{t_0,x_0}}\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\int_a^b |Z_s^{a,x}|^2 ds\right]_{|x=X_a^{t_0,x_0}}\right].
\end{aligned}$$

Now, for any $x \in \mathbb{R}^d$, one has $Z_s^{b,x} = 0$ if $s \leq b$, by construction. Hence

$$\mathbb{E}\left[\int_a^b |Z_s^{a,x}|^2 ds\right] = \mathbb{E}\left[\int_a^b |Z_s^{a,x} - Z_s^{b,x}|^2 ds\right] \leq \|Z^{a,x} - Z^{b,x}\|_{\mathcal{H}^2}^2 \leq C(1 + |x|^2)(b - a),$$

by (26) again. Therefore

$$\mathbb{E}\left[\int_a^b |Z_s^{t_0,x_0}|^2 ds\right] \leq \mathbb{E}\left[C(1 + |X_a^{t_0,x_0}|^2)|a - b|\right] \leq C(1 + |x_0|^2)(b - a).$$

Now, if $T < a < b \leq T + \theta$, we have $Z_r^{t_0,x_0} = \Psi(r, X_r^{t_0,x_0})$ for $r \in [a, b]$, hence

$$\mathbb{E}\left[\int_a^b |Z_r^{t_0,x_0}|^2 dr\right] = \mathbb{E}\left[\int_a^b |\Psi(r, X_r^{t_0,x_0})|^2 dr\right] \leq C(1 + |x_0|^2)(b - a).$$

Finally, if $a \leq T < b \leq T + \theta$, the result follows by writing

$$\mathbb{E}\left[\int_a^b |Z_r^{t_0,x_0}|^2 dr\right] = \mathbb{E}\left[\int_a^T |Z_r^{t_0,x_0}|^2 dr\right] + \mathbb{E}\left[\int_T^b |Z_r^{t_0,x_0}|^2 dr\right]$$

and applying the previous cases. We are done. □

Remark 4.1. The case $m_t(Y, Z) = \mathcal{L}_t(Z_{t+\theta})$ is not covered by Lemma 4.1. Indeed, the above derivations exploit the Hölder regularity of the Y component of the Markovian ABSDE solution, which is not guaranteed regarding the Z process. Hence the results of this section do not apply to this case.

For x fixed in $\mathbb{R}^d$, we consider the solution (X, Y, Z) to (21) with initial condition $(0, \mathrm{x})$ of X. Let $\widetilde{m}$ be the map introduced in Assumption 4.1(iv). Since $\widetilde{m}$ is continuous and X is continuous and adapted, $(\widetilde{m}(t, X_t))_{t\in[0,T]}$ is progressive and a version of $m(Y, Z)$. Throughout this section, we identify $m(Y, Z)$ with this version.

Remark 4.2. A priori, $m(Y, Z)$ is only defined $dt \otimes d\mathbb{P}$-a.e. Here, in order to consider discrete schemes, it has to be evaluated at deterministic grid times, hence the need for choosing a (measurable) representative of it.

4.2 Zhang's Lemma

Given integers $N' \geq N \geq 1$, we fix a time-grid $\{0 = t_0 < \cdots < t_N = T < t_{N+1} < \cdots < t_{N'} = T + \theta\}$ of mesh size $\mathrm{h} = \max_{0 \leq n < N'} \mathrm{h}_n$, where $\mathrm{h}_n = t_{n+1} - t_n$, and we set $I_m = [t_m, t_{m+1})$, $m < N' - 1$, and $I_{N'-1} = [t_{N'-1}, t_{N'}]$.

Defining

$$\widetilde{Z}^{\mathrm{h}}_{t_n} = \mathrm{h}_n^{-1} \mathbb{E}_{t_n} \left[\int_{t_n}^{t_{n+1}} Z_s ds \right], \quad 0 \leq n < N', \tag{33}$$

we have the following result (Zhang (2001)).

Lemma 4.2. *Under Assumptions 3.1, 3.2 and 4.1, there exists a constant $C \geq 0$ such that,*

$$\sum_{n=0}^{N'-1} \int_{t_n}^{t_{n+1}} \mathbb{E} \left[|Z_t - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2 \right] dt \leq C\mathrm{h}.$$

Proof. By Assumption 4.1(iv), (Y, Z) is the solution to the classical BSDE

$$Y_t = \Phi(T, X_T) + \int_t^T f(s, X_s, Y_s, Z_s, \widetilde{m}(s, X_s)) ds - \int_t^T Z_s dW_s, \quad 0 \leq t \leq T,$$

From Zhang (2001, Theorem 3.4.3) (which proof still applies under the local Hölder continuity induced Assumption 4.1(iv) as $\sup_{t \leq T} \mathbb{E}\left[|X_t|^2\right] < \infty$), we have

$$\sum_{n=0}^{N-1} \int_{t_n}^{t_{n+1}} \mathbb{E} \left[|Z_t - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2 \right] dt \leq C\mathrm{h}.$$

It remains to control the sum over $N \leq n < N'$, which is easily done as follows, since $Z = \Psi(\cdot, X)$ on $(T, T+\theta]$ (and we choose a version of Z such that we also have $Z_T = \Psi(T, X_T)$). By Jensen's inequality and using Assumptions 3.1 and 4.1(i), we have

$$\begin{aligned}
&\sum_{n=N}^{N'-1} \int_{t_n}^{t_{n+1}} \mathbb{E} \left[|Z_t - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2 \right] dt \\
\leq & 2 \sum_{n=N}^{N'-1} \int_{t_n}^{t_{n+1}} \mathbb{E} \left[|Z_t - Z_{t_n}|^2 \right] dt + 2 \sum_{n=N}^{N'-1} \mathrm{h}_n \mathbb{E} \left[|Z_{t_n} - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2 \right] \\
\leq & 2 \sum_{n=N}^{N'-1} \int_{t_n}^{t_{n+1}} \mathbb{E} \left[|\Psi(t, X_t) - \Psi(t_n, X_{t_n})|^2 \right] dt \\
& + 2 \sum_{n=N}^{N'-1} \mathrm{h}_n \mathbb{E} \left[|\Psi(t_n, X_{t_n}) - \mathrm{h}_n^{-1} \mathbb{E}_{t_n} \left[\int_{t_n}^{t_{n+1}} \Psi(s, X_s) ds \right] |^2 \right] \\
\leq & C\mathrm{h} + 2 \sum_{n=N}^{N'-1} \mathrm{h}_n \mathbb{E} \left[|\mathbb{E}_{t_n} \left[\int_{t_n}^{t_{n+1}} \Psi(t_n, X_{t_n}) - \Psi(s, X_s) \frac{ds}{\mathrm{h}_n} \right] |^2 \right] \\
& \leq C\mathrm{h} + 2 \sum_{n=N}^{N'-1} \int_{t_n}^{t_{n+1}} \mathbb{E} \left[|\Psi(t_n, X_{t_n}) - \Psi(s, X_s)|^2 \right] ds \\
& \leq C\mathrm{h},
\end{aligned}$$

where $C \geq 0$ is a constant that can change from line to line (but is independent of h). □

4.3 The Discrete-Time Scheme

In order to discretise in time the Markovian ABSDE, we first consider X^{h} defined by the (continuous-time version of the) Euler-Maruyama scheme X^{h} of X, defined by $X^{\mathrm{h}}_{t_0} = \mathrm{x}$ and, for $t_n \le t \le t_{n+1}, 0 \le n < N'$,

$$X^{\mathrm{h}}_t = X^{\mathrm{h}}_{t_n} + b(t_n, X^{\mathrm{h}}_{t_n})(t - t_n) + \sigma(t_n, X^{\mathrm{h}}_{t_n})(W_t - W_{t_n}).$$

By Kloeden and Platen (1992, Theorem 10.2.2), under Assumption 3.1(iv),

$$\mathbb{E}\left[\sup_{0\le t\le T+\theta} |X^{\mathrm{h}}_t - X_t|^2\right] \le C\mathrm{h}. \tag{34}$$

Under Assumption 3.1(i), we readily obtain the following.

Lemma 4.3. *Under Assumption 3.1(i) and (iv), we have*

$$\mathbb{E}\left[\sup_{T\le t\le T+\theta} |\Phi(t, X^{\mathrm{h}}_t) - \Phi(t, X_t)|^2\right] + \mathbb{E}\left[\sup_{T< t\le T+\theta} |\Psi(t, X^{\mathrm{h}}_t) - \Psi(t, X_t)|^2\right] \le C\Lambda^2\mathrm{h}.$$

Regarding the ABSDE, we introduce the following scheme: $Y^{\mathrm{h}}_{t_n} = \Phi(t_n, X^{\mathrm{h}}_{t_n})$ and $Z^{\mathrm{h}}_{t_n} = \Psi(t_n, X^{\mathrm{h}}_{t_n})$, $N \le n \le N'$, and, for $0 \le n < N$,

$$\begin{aligned} Z^{\mathrm{h}}_{t_n} &= \mathrm{h}_n^{-1}\mathbb{E}_{t_n}\left[(W_{t_{n+1}} - W_{t_n})Y^{\mathrm{h}}_{t_{n+1}}\right], \\ Y^{\mathrm{h}}_{t_n} &= \mathbb{E}_{t_n}\left[Y^{\mathrm{h}}_{t_{n+1}}\right] + f(t_n, X^{\mathrm{h}}_{t_n}, Y^{\mathrm{h}}_{t_n}, Z^{\mathrm{h}}_{t_n}, m^{\mathrm{h}}_{t_n})\mathrm{h}_n, \end{aligned}$$

where, for $0 \le n < N$, $m^{\mathrm{h}}_{t_n}$ is a $\mathcal{P}_2(E)$ valued, $\mathfrak{F}_{t_n}$ measurable random variable, approximating the anticipated term.

At the n-th step of the backward recursion, before computing $Y^{\mathrm{h}}_{t_n}$, the variables

$$(Y^{\mathrm{h}}_{t_k})_{k>n}, \qquad (Z^{\mathrm{h}}_{t_k})_{k\ge n},$$

as well as X^{h}, are already available, since $Z^{\mathrm{h}}_{t_n}$ is computed before $Y^{\mathrm{h}}_{t_n}$. With these quantities held fixed, we assume that

$$m^{\mathrm{h}}_{t_n} = \mathcal{M}^{\mathrm{h}}_{t_n}(Y^{\mathrm{h}}_{t_n}),$$

for some map $\mathcal{M}^{\mathrm{h}}_{t_n} : L^2(\mathfrak{F}_{t_n}) \to L^2(\mathfrak{F}_{t_n}, \mathcal{P}_2(E))$, where the dependence of $\mathcal{M}^{\mathrm{h}}_{t_n}$ on the quantities already available before computing $Y^{\mathrm{h}}_{t_n}$ is suppressed from the notation.

In order to have a well-defined scheme, we introduce the following assumption on $\mathcal{M}^{\mathrm{h}}_{t_n}$.

Assumption 4.2. We assume that for all $\chi, \chi' \in L^2(\mathfrak{F}_{t_n})$,

$$\mathbb{E}\left[W_2^2(\mathcal{M}^{\mathrm{h}}_{t_n}(\chi), \mathcal{M}^{\mathrm{h}}_{t_n}(\chi'))\right] \le \Lambda\mathbb{E}\left[|\chi - \chi'|^2\right]. \tag{35}$$

Lemma 4.4. *Under Assumptions 3.1 and 4.2, for any grid and $0 \le n \le N'$, $Y^{\mathrm{h}}_{t_n}$ is uniquely defined in $L^2(\mathfrak{F}_{t_n}, \mathbb{R})$, as soon as $\mathrm{h}^2\Lambda^2(1+\Lambda) < \frac{1}{2}$, and $Z^{\mathrm{h}}_{t_n}$ is uniquely defined in $L^2(\mathfrak{F}_{t_n}, \mathbb{R}^q)$.*

Proof. Fix $0 \le n < N$ and suppose by backward induction that the future grid values $(Y^{\mathrm{h}}_{t_k}, Z^{\mathrm{h}}_{t_k})_{k>n}$ have been constructed. After computing $Z^{\mathrm{h}}_{t_n}$, keep these quantities fixed and consider the map

$$\chi \mapsto \mathbb{E}_{t_n}\left[Y^{\mathrm{h}}_{t_{n+1}}\right] + f(t_n, X^{\mathrm{h}}_{t_n}, \chi, Z^{\mathrm{h}}_{t_n}, \mathcal{M}^{\mathrm{h}}_{t_n}(\chi))\mathrm{h}_n.$$

Then this map is a contraction of $L^2(\mathfrak{F}_{t_n}, \mathbb{R})$ whenever $\Lambda^2 \mathrm{h}^2(1+\Lambda) < \frac{1}{2}$, as

$$\begin{aligned}
&\mathbb{E}\left[|f(t_n, X^{\mathrm{h}}_{t_n}, \chi, Z^{\mathrm{h}}_{t_n}, \mathcal{M}^{\mathrm{h}}_{t_n}(\chi))\mathrm{h}_n - f(t_n, X^{\mathrm{h}}_{t_n}, \chi', Z^{\mathrm{h}}_{t_n}, \mathcal{M}^{\mathrm{h}}_{t_n}(\chi'))\mathrm{h}_n|^2\right] \\
&\leq 2\Lambda^2 \mathrm{h}_n^2 \mathbb{E}\left[|\chi - \chi'|^2 + W_2^2(\mathcal{M}^{\mathrm{h}}_{t_n}(\chi), \mathcal{M}^{\mathrm{h}}_{t_n}(\chi'))\right] \\
&\leq 2\Lambda^2 \mathrm{h}^2(1+\Lambda)\mathbb{E}\left[|\chi - \chi'|^2\right].
\end{aligned}$$

Hence it has a unique fix-point $Y^{\mathrm{h}}_{t_n}$.

Regarding $Z^{\mathrm{h}}_{t_n}$, the conditional Cauchy-Schwarz inequality yields

$$\mathbb{E}\left[|Z^{\mathrm{h}}_{t_n}|^2\right] \leq q\mathrm{h}_n^{-1}\mathbb{E}\left[|Y^{\mathrm{h}}_{t_{n+1}}|^2\right] < \infty.$$

□

4.4 Continuous-Time Interpolated Scheme

Now, assuming that the discrete scheme well-defined, e.g. under Assumption 3.1 and 4.2 by the previous lemma, using the Brownian martingale representation property, for each $0 \leq n < N$, there exists a square-integrable process $\overline{Z}^{\mathrm{h}}$ such that

$$\begin{aligned}
Y^{\mathrm{h}}_{t_n} &= \mathbb{E}_{t_n}\left[Y^{\mathrm{h}}_{t_{n+1}}\right] + \mathrm{h}_n f(t_n, X^{\mathrm{h}}_{t_n}, Y^{\mathrm{h}}_{t_n}, Z^{\mathrm{h}}_{t_n}, m^{\mathrm{h}}_{t_n}) && (36)\\
&= Y^{\mathrm{h}}_{t_{n+1}} + \mathrm{h}_n f(t_n, X^{\mathrm{h}}_{t_n}, Y^{\mathrm{h}}_{t_n}, Z^{\mathrm{h}}_{t_n}, m^{\mathrm{h}}_{t_n}) - \int_{t_n}^{t_{n+1}} \overline{Z}^{\mathrm{h}}_s dW_s. && (37)
\end{aligned}$$

We then define, for each $0 \leq n < N$ and $t_n \leq t \leq t_{n+1}$,

$$\overline{Y}^{\mathrm{h}}_t = Y^{\mathrm{h}}_{t_n} - f(t_n, X^{\mathrm{h}}_{t_n}, Y^{\mathrm{h}}_{t_n}, Z^{\mathrm{h}}_{t_n}, m^{\mathrm{h}}_{t_n})(t - t_n) + \int_{t_n}^{t} \overline{Z}^{\mathrm{h}}_s dW_s. \tag{38}$$

Multiplying by $(W_{t_{n+1}} - W_{t_n})$ and taking conditional expectation, we obtain by Itô isometry

$$\mathbb{E}_{t_n}\left[Y^{\mathrm{h}}_{t_{n+1}}(W_{t_{n+1}} - W_{t_n})\right] = \mathbb{E}_{t_n}\left[\int_{t_n}^{t_{n+1}} \overline{Z}^{\mathrm{h}}_s ds\right],$$

i.e., dividing by h_n,

$$Z^{\mathrm{h}}_{t_n} = \mathrm{h}_n^{-1}\mathbb{E}_{t_n}\left[\int_{t_n}^{t_{n+1}} \overline{Z}^{\mathrm{h}}_s ds\right]. \tag{39}$$

We also define $\overline{Y}^{\mathrm{h}}_s = \Phi(s, X^{\mathrm{h}}_s)$ and $\overline{Z}^{\mathrm{h}}_s = \Psi(s, X^{\mathrm{h}}_s)$ for $s \in (T, T+\theta]$.

4.5 Preparatory Results

Now, in order to prove convergence of the discrete scheme, we need the following consistency assumption regarding the anticipated term.

Assumption 4.3. There exists $C \geq 0$, independent of the grid and of $n \in \{0, \dots, N'\}$, such that, for any $0 \leq n < N$,

$$\mathbb{E}\left[W_2^2(m^{\mathrm{h}}_{t_n}, \widetilde{m}(t_n, X_{t_n}))\right] \leq C\left(\mathrm{h} + \max_{n \leq k \leq N} \mathbb{E}\left[|Y^{\mathrm{h}}_{t_k} - Y_{t_k}|^2\right] + \mathbb{E}\left[\int_{t_n}^{T} |Z_t - \overline{Z}^{\mathrm{h}}_t|^2 dt\right]\right). \tag{40}$$

The following lemma shows that this is satisfied in the two pilot cases of Lemma 4.1.

Lemma 4.5. *Let Assumptions 3.1 and 3.2 hold.*
(i) *Under Assumption 4.1(i), assume that* $m_t(Y,Z) = \mathcal{L}_t(Y_{t+\theta})$ *for any* $(Y,Z) \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ *and* $t \in [0,T]$, *and* $m^{\mathrm{h}}_{t_n} = \mathcal{L}_{t_n}(Y^{\mathrm{h}}_{t_{\kappa(n)}})$, *where* $\kappa(n) = \inf\{k \in \{n,\dots,N'\} | t_n + \theta \le t_k\}$. *Then Assumptions 4.2 and 4.3 hold.*
(ii) *Under Assumption 4.1, assume that* $t_n + \theta$ *is a grid point for any* $0 \le n < N$. *Assume* $m_t(Y,Z) = \mathcal{L}_t(\int_t^{t+\theta} Z_s dW_s)$ *for any* $(Y,Z) \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ *and* $t \in [0,T]$, *and consider* $m^{\mathrm{h}}_{t_n} = \mathcal{L}_{t_n}(\sum_{k=n}^{\kappa(n)-1} Z^{\mathrm{h}}_{t_k}(W_{t_{k+1}} - W_{t_k}))$, *where* $\kappa(n)$ *is such that* $t_n + \theta = t_{\kappa(n)}$. *Then, Assumptions 4.2 and 4.3 hold.*

Proof. **(i)** *(Assumption 4.2.)* If $m_t(Y,Z) = \mathcal{L}_t(Y_{t+\theta})$ and $\theta > 0$, then $\kappa(n) > n$, hence $\mathcal{M}^{\mathrm{h}}(\chi) = \mathcal{L}_{t_n}(Y^{\mathrm{h}}_{t_{\kappa(n)}})$ does not depend on χ, so that (35) is trivially satisfied. Otherwise, $\theta = 0$,so $m_t(Y,Z) = \delta_{Y_t}$ and $\mathcal{M}^{\mathrm{h}}_{t_n}(\chi) = \delta_\chi$, so $\mathbb{E}\left[W_2^2(\mathcal{M}^{\mathrm{h}}_{t_n}(\chi), \mathcal{M}^{\mathrm{h}}_{t_n}(\chi'))\right] = \mathbb{E}\left[|\chi - \chi'|^2\right]$ and (35) is satisfied.

(Assumption 4.3.) Regarding (40), we compute

$$\begin{aligned}\mathbb{E}\left[W_2^2(m^{\mathrm{h}}_{t_n}, \widetilde{m}(t_n, X_{t_n}))\right] &= \mathbb{E}\left[W_2^2(\mathcal{L}_{t_n}(Y^{\mathrm{h}}_{t_{\kappa(n)}}), \mathcal{L}_{t_n}(Y_{t_n+\theta}))\right] \\ &\le \mathbb{E}\left[|Y^{\mathrm{h}}_{t_{\kappa(n)}} - Y_{t_n+\theta}|^2\right] \\ &\le 2\mathbb{E}\left[|Y^{\mathrm{h}}_{t_{\kappa(n)}} - Y_{t_{\kappa(n)}}|^2\right] + 2\mathbb{E}\left[|Y_{t_{\kappa(n)}} - Y_{t_n+\theta}|^2\right].\end{aligned}$$

If $\kappa(n) \le N$, Theorem 3.6 gives $\mathbb{E}\left[|Y_{t_{\kappa(n)}} - Y_{t_n+\theta}|^2\right] \le C\mathrm{h}$, so we obtain

$$\mathbb{E}\left[W_2^2(m^{\mathrm{h}}_{t_n}, \widetilde{m}(t_n, X_{t_n}))\right] \le C\left(\max_{n\le k\le N} \mathbb{E}\left[|Y^{\mathrm{h}}_{t_k} - Y_{t_k}|^2\right] + \mathrm{h}\right),$$

which implies (40). Otherwise, $\kappa(n) > N$, and Assumption 4.1(i) implie, by definition of $\kappa(n)$, $\mathbb{E}\left[|Y_{t_{\kappa(n)}} - Y_{t_n+\theta}|^2\right] \le C|t_{\kappa(n)} - (t_n+\theta)| \le C|t_{\kappa(n)} - t_{\kappa(n)-1}| \le C\mathrm{h}$ and we have, using also (34),

$$\mathbb{E}\left[W_2^2(m^{\mathrm{h}}_{t_n}, \widetilde{m}(t_n, X_{t_n}))\right] \le C(\mathbb{E}\left[|\Phi(t_{\kappa(n)}, X^{\mathrm{h}}_{t_{\kappa(n)}}) - \Phi(t_{\kappa(n)}, X_{t_{\kappa(n)}})|^2\right] + \mathrm{h}) \le C\mathrm{h},$$

which implies again (40).
(ii) *(Assumption 4.2.)* As $m^{\mathrm{h}}_{t_n}$ does not depend on $Y^{\mathrm{h}}_{t_n}$, (35) trivially holds.

(Assumption 4.3.) Regarding (40), we compute, with $\widetilde{Z}^{\mathrm{h}}$ introduced in (33),

$$\begin{aligned}\mathbb{E}\left[W_2^2(m^{\mathrm{h}}_{t_n}, \widetilde{m}(t_n, X_{t_n}))\right] &\le \mathbb{E}\left[\Big|\int_{t_n}^{t_n+\theta} Z_s dW_s - \sum_{k=n}^{\kappa(n)-1} Z^{\mathrm{h}}_{t_k}(W_{t_{k+1}} - W_{t_k})\Big|^2\right] \\ &= \mathbb{E}\left[\sum_{k=n}^{\kappa(n)-1} \int_{t_k}^{t_{k+1}} |Z_s - Z^{\mathrm{h}}_{t_k}|^2 ds\right] \le \mathbb{E}\left[\sum_{k=n}^{N'-1} \int_{t_k}^{t_{k+1}} |Z_s - Z^{\mathrm{h}}_{t_k}|^2 ds\right] \\ &\le 2\mathbb{E}\left[\sum_{k=n}^{N-1} \int_{t_k}^{t_{k+1}} |Z_s - \widetilde{Z}^{\mathrm{h}}_{t_k}|^2 ds\right] + 2\mathbb{E}\left[\sum_{k=n}^{N-1} \mathrm{h}_k |\widetilde{Z}^{\mathrm{h}}_{t_k} - Z^{\mathrm{h}}_{t_k}|^2\right] \\ &\quad + \mathbb{E}\left[\sum_{k=N}^{N'-1} \int_{t_k}^{t_{k+1}} |Z_s - Z^{\mathrm{h}}_{t_k}|^2 ds\right].\end{aligned}$$

From Lemma 4.2, the first term is upper bounded by $C\mathrm{h}$, while for the second term, we compute for each $n \le k \le N-1$,

$$\mathbb{E}\left[|\widetilde{Z}^{\mathrm{h}}_{t_k} - Z^{\mathrm{h}}_{t_k}|^2\right] = \mathbb{E}\left[\left|\mathbb{E}_{t_k}\left[\mathrm{h}_k^{-1}\int_{t_k}^{t_{k+1}} (Z_s - \overline{Z}^{\mathrm{h}}_s) ds\right]\right|^2\right] \le \mathrm{h}_k^{-1}\mathbb{E}\left[\int_{t_k}^{t_{k+1}} |Z_s - \overline{Z}^{\mathrm{h}}_s|^2 ds\right].$$

Regarding the last term, we compute

$$\sum_{k=N}^{N'-1}\int_{t_k}^{t_{k+1}}\mathbb{E}\left[|\Psi(s,X_s)-\Psi(t_k,X^{\mathrm{h}}_{t_k})|^2\right]ds\le C\sum_{k=N}^{N'-1}\int_{t_k}^{t_{k+1}}(\mathrm{h}+\mathbb{E}\left[|X_s-X^{\mathrm{h}}_{t_k}|^2\right])ds\le C\mathrm{h}.$$

Hence

$$\mathbb{E}\left[W_2^2(m^{\mathrm{h}}_{t_n},\widetilde{m}(t_n,X_{t_n}))\right]\le C\mathrm{h}+2\int_{t_n}^{T}\mathbb{E}\left[|Z_s-\overline{Z}^{\mathrm{h}}_s|^2\right]ds,$$

which proves (40). □

In order to control the error between (Y,Z) and $(\overline{Y}^{\mathrm{h}},\overline{Z}^{\mathrm{h}})$, we also need the following generalization of the classical discrete Gronwall's lemma.

Lemma 4.6. *Let $(e_n)_{n=0}^N$, $(f_n)_{n=0}^{N-1}$ and $(\varepsilon_n)_{n=0}^{N-1}$ be sequences of non-negative real numbers. Suppose there exists a constant $G\ge 0$ such that*

$$e_n+\frac{1}{2}f_n\le(1+G\mathrm{h}_n)(e_{n+1}+\mathrm{h}_nE_n+\mathrm{h}_nF_n)+\varepsilon_n,,\ 0\le n\le N-1,$$

where $E_n=\max_{n\le k\le N}e_k$ and $F_n=\sum_{k=n}^{N-1}f_k$ (and $F_N=0$).
Then, assuming $\mathrm{h}\le h_0(G)$, where $h_0(G)>0$ is such that $3(1+Gh_0)h_0\le\frac{1}{2}$,

$$E_0+F_0\le C(G,T)\left(e_N+\sum_{k=0}^{N-1}\varepsilon_k\right).$$

Proof. First, define, for $0\le n\le N$, $A_n=e_n+\frac{1}{2}F_n$ and $B_n=\max_{n\le k\le N}A_k$, and observe that for fixed n, for $n\le k\le N$, $e_k\le A_k\le B_n$, so $e_n\le E_n\le B_n$. One also has $F_n=2A_n-2e_n\le 2A_n\le 2B_n$. One then computes

$$\begin{aligned}A_n&=e_n+\frac{1}{2}f_n+\frac{1}{2}F_{n+1}\\&\le(1+G\mathrm{h}_n)\,e_{n+1}+\frac{1}{2}\,F_{n+1}+(1+G\mathrm{h}_n)\,\mathrm{h}_n\,(E_n+F_n)+\varepsilon_n\\&\le e_{n+1}+\frac{1}{2}F_{n+1}+G\mathrm{h}_ne_{n+1}+3(1+G\mathrm{h}_n)\mathrm{h}_nB_n+\varepsilon_n\\&\le(1+G\mathrm{h}_n)B_{n+1}+3(1+G\mathrm{h}_n)\mathrm{h}_nB_n+\varepsilon_n.\end{aligned}$$

Since $B_n=\max(A_n,B_{n+1})$, we also have

$$B_n\le(1+G\mathrm{h}_n)B_{n+1}+3(1+G\mathrm{h}_n)\mathrm{h}_nB_n+\varepsilon_n,$$

hence

$$(1-3(1+G\mathrm{h}_n)\mathrm{h}_n)B_n\le(1+G\mathrm{h}_n)B_{n+1}+\varepsilon_n.$$

Since h is such that $3(1+G)\mathrm{h}\le\frac{1}{2}$, we get

$$B_n\le(1+G_1\mathrm{h}_n)B_{n+1}+2\varepsilon_n,$$

where $G_1:=2(G+3+3Gh_0(G))$.

From the classical discrete Gronwall's lemma, we then obtain, as $e_N=A_N=B_N$, that

$$B_0\le e^{G_1T}\left(e_N+2\sum_{j=0}^{N-1}\varepsilon_j\right).$$

Last, as $E_0\le B_0$ and $F_0\le 2A_0\le 2B_0$, we obtain

$$E_0+F_0\le 3B_0\le 3e^{G_1T}\left(e_N+2\sum_{j=0}^{N-1}\varepsilon_j\right),$$

which gives the result for $C(G,T)=6\,e^{G_1T}$. □

4.6 Convergence Analysis

Theorem 4.7. *Under Assumptions 3.1, 3.2, 4.1, 4.2 and 4.3, there exists a constant $C \geq 0$ (independent of* h*) such that, for sufficiently small* h*,*

$$\sup_{t\in[0,T+\theta]} \mathbb{E}\left[|Y_t - \overline{Y}_t^{\mathrm{h}}|^2\right] + \mathbb{E}\left[\int_0^{T+\theta} |Z_t - \overline{Z}_t^{\mathrm{h}}|^2\, dt\right] \leq C\,\mathrm{h}.$$

Proof. (Local error estimate.) For $0 \leq n < N$ and $t_n \leq t < t_{n+1}$, set

$$\begin{aligned}
&\delta Y_t = Y_t - Y_t^{\mathrm{h}}, \qquad \delta Z_t = Z_t - \overline{Z}_t^{\mathrm{h}},\\
&\delta f_t = f(t, X_t, Y_t, Z_t, \widetilde{m}(t, X_t)) - f(t_n, X_{t_n}^{\mathrm{h}}, Y_{t_n}^{\mathrm{h}}, Z_{t_n}^{\mathrm{h}}, m_{t_n}^{\mathrm{h}}).
\end{aligned}$$

For each $0 \leq n < N$ and $t_n \leq t < t_{n+1}$, applying Itô's formula to $|\delta Y_t|^2$ yields

$$\mathbb{E}\left[|\delta Y_t|^2\right] + \mathbb{E}\left[\int_t^{t_{n+1}} |\delta Z_s|^2 ds\right] = \mathbb{E}\left[|\delta Y_{t_{n+1}}|^2\right] + 2\mathbb{E}\left[\int_t^{t_{n+1}} \delta Y_s^\top\, \delta f_s\, ds\right].$$

Using Young's inequality $2ab \leq \alpha a^2 + \alpha^{-1} b^2$ for a parameter $\alpha > 0$ to be chosen later,

$$\begin{aligned}
&\mathbb{E}\left[|\delta Y_t|^2\right] + \mathbb{E}\left[\int_t^{t_{n+1}} |\delta Z_s|^2 ds\right] \\
&\qquad \leq \mathbb{E}\left[|\delta Y_{t_{n+1}}|^2\right] + \alpha\mathbb{E}\left[\int_t^{t_{n+1}} |\delta Y_s|^2 ds\right] + \frac{1}{\alpha}\mathbb{E}\left[\int_t^{t_{n+1}} |\delta f_s|^2 ds\right].
\end{aligned} \tag{41}$$

By the Lipschitz property of f in (x, y, z, μ) and its Hölder regularity in time (see Assumptions 3.1 and 4.1(iii)), for $s \in [t_n, t_{n+1})$,

$$\begin{aligned}
|\delta f_s|^2 &\leq C\Big(|s - t_n| + |X_s - X_{t_n}^{\mathrm{h}}|^2 + |Y_s - Y_{t_n}^{\mathrm{h}}|^2 + |Z_s - Z_{t_n}^{\mathrm{h}}|^2 + W_2^2(\widetilde{m}(s, X_s), m_{t_n}^{\mathrm{h}})\Big)\\
&\leq C\Big(\mathrm{h} + |X_s - X_{t_n}^{\mathrm{h}}|^2 + |Y_s - Y_{t_n}|^2 + |\delta Y_{t_n}|^2\\
&\quad + |Z_s - \widetilde{Z}_{t_n}^{\mathrm{h}}|^2 + |\widetilde{Z}_{t_n}^{\mathrm{h}} - Z_{t_n}^{\mathrm{h}}|^2\\
&\quad + W_2^2(\widetilde{m}(s, X_s), \widetilde{m}(t_n, X_{t_n})) + W_2^2(\widetilde{m}(t_n, X_{t_n}), m_{t_n}^{\mathrm{h}})\Big),
\end{aligned}$$

where $\widetilde{Z}_{t_n}^{\mathrm{h}}$ is defined in (33).

For any $t_n \leq s \leq t_{n+1}$, we have, from (34) and from Theorem 3.6, for some constant $C \geq 0$,

$$\begin{aligned}
\mathbb{E}\left[|X_s - X_{t_n}^{\mathrm{h}}|^2\right] &\leq 2\mathbb{E}\left[|X_s - X_{t_n}|^2\right] + 2\mathbb{E}\left[|X_{t_n} - X_{t_n}^{\mathrm{h}}|^2\right] \leq C\mathrm{h},\\
\mathbb{E}\left[|Y_s - Y_{t_n}|^2\right] &= \mathbb{E}\left[|u(s, X_s) - u(t_n, X_{t_n})|^2\right] \leq C\mathrm{h}.
\end{aligned}$$

In addition, from (39) and the definition of $\widetilde{Z}_{t_n}^{\mathrm{h}}$, by Jensen's inequality, we get

$$\mathbb{E}\left[|\widetilde{Z}_{t_n}^{\mathrm{h}} - Z_{t_n}^{\mathrm{h}}|^2\right] \leq \mathbb{E}\left[\left|\mathbb{E}_{t_n}\left[\mathrm{h}_n^{-1}\int_{t_n}^{t_{n+1}} \delta Z_s ds\right]\right|^2\right] \leq \mathrm{h}_n^{-1}\mathbb{E}\left[\int_{t_n}^{t_{n+1}} |\delta Z_s|^2 ds\right]. \tag{42}$$

Last, the anticipated terms are controlled by Assumptions 4.1(iv) and 4.3:

$$\begin{aligned}
&\mathbb{E}\left[W_2^2(\widetilde{m}(s, X_s), \widetilde{m}(t_n, X_{t_n}))\right] + \mathbb{E}\left[W_2^2(m_{t_n}^{\mathrm{h}}, \widetilde{m}(t_n, X_{t_n}))\right]\\
\leq& C\mathrm{h} + C\max_{n\leq k\leq N}\mathbb{E}\left[|\delta Y_{t_k}|^2\right] + C\mathbb{E}\left[\int_{t_n}^{T} |\delta Z_t|^2 dt\right].
\end{aligned}$$

All in all, we have shown that, for any $t_n \le s \le t_{n+1}$,

$$\begin{aligned}&\mathbb{E}\left[|\delta f_s|^2\right]\\ \le& C\Big(\mathrm{h} + \max_{n\le k\le N}\mathbb{E}\left[|\delta Y_{t_k}|^2\right] + \mathbb{E}\left[|Z_s - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2\right] + \mathrm{h}_n^{-1}\int_{t_n}^{t_{n+1}}\mathbb{E}\left[|\delta Z_u|^2\right]du + \int_{t_n}^{T}\mathbb{E}\left[|\delta Z_u|^2\right]du\Big),\end{aligned}$$

hence, for any $t_n \le t < t_{n+1}$,

$$\begin{aligned}\mathbb{E}\left[|\delta Y_t|^2\right] + \mathbb{E}\left[\int_t^{t_{n+1}}|\delta Z_s|^2 ds\right] \le& \mathbb{E}\left[|\delta Y_{t_{n+1}}|^2\right] + \alpha\mathbb{E}\left[\int_t^{t_{n+1}}|\delta Y_s|^2 ds\right]\\ &+ \frac{C}{\alpha}\Big(\mathrm{h}_n\mathrm{h} + \mathrm{h}_n\max_{n\le k\le N}\mathbb{E}\left[|\delta Y_{t_k}|^2\right] + \int_{t_n}^{t_{n+1}}\mathbb{E}\left[|Z_s - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2\right]ds\\ &+ \int_{t_n}^{t_{n+1}}\mathbb{E}\left[|\delta Z_s|^2\right]ds + \mathrm{h}_n\int_{t_n}^{T}\mathbb{E}\left[|\delta Z_s|^2\right]ds\Big).\end{aligned}$$

(Discrete error recursion.) Using Gronwall's lemma on the interval $[t_n, t_{n+1}]$, we obtain, for each $t_n \le t \le t_{n+1}$,

$$\begin{aligned}\mathbb{E}\left[|\delta Y_t|^2\right] \le e^{\alpha\mathrm{h}_n}\Big(&\mathbb{E}\left[|\delta Y_{t_{n+1}}|^2\right] + \frac{C}{\alpha}\Big(\mathrm{h}_n\mathrm{h} + \mathrm{h}_n\max_{n\le k\le N}\mathbb{E}\left[|\delta Y_{t_k}|^2\right]\\ &+ \int_{t_n}^{t_{n+1}}\mathbb{E}\left[|Z_s - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2\right]ds + \int_{t_n}^{t_{n+1}}\mathbb{E}\left[|\delta Z_s|^2\right]ds + \mathrm{h}_n\int_{t_n}^{T}\mathbb{E}\left[|\delta Z_s|^2\right]ds\Big)\Big),\end{aligned} \tag{43}$$

and substituting this estimate back into the previous inequality to get

$$\begin{aligned}&\mathbb{E}\left[|\delta Y_t|^2\right] + \mathbb{E}\left[\int_t^{t_{n+1}}|\delta Z_s|^2 ds\right]\\ \le&(1+\alpha\mathrm{h}_n e^{\alpha\mathrm{h}_n})\Big(\mathbb{E}\left[|\delta Y_{t_{n+1}}|^2\right] + \frac{C}{\alpha}\Big(\mathrm{h}_n\mathrm{h} + \mathrm{h}_n\max_{n\le k\le N}\mathbb{E}\left[|\delta Y_{t_k}|^2\right]\\ &+ \int_{t_n}^{t_{n+1}}\mathbb{E}\left[|Z_s - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2\right]ds + \int_{t_n}^{t_{n+1}}\mathbb{E}\left[|\delta Z_s|^2\right]ds + \mathrm{h}_n\int_{t_n}^{T}\mathbb{E}\left[|\delta Z_s|^2\right]ds\Big)\Big).\end{aligned} \tag{44}$$

Choosing α large enough to absorb the penultimate term $\int_{t_n}^{t_{n+1}}\mathbb{E}\left[|\delta Z_s|^2\right]ds$ in the left hand side, and then h small enough, we obtain, for $t = t_n$,

$$\begin{aligned}\mathbb{E}\left[|\delta Y_{t_n}|^2\right] + \frac{1}{2}\mathbb{E}\left[\int_{t_n}^{t_{n+1}}|\delta Z_s|^2 ds\right] \le&(1+C\mathrm{h}_n)\Big(\mathbb{E}\left[|\delta Y_{t_{n+1}}|^2\right] + \mathrm{h}_n\mathrm{h} + \int_{t_n}^{t_{n+1}}\mathbb{E}\left[|Z_s - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2\right]ds\\ &+ \mathrm{h}_n\max_{n\le k\le N}\mathbb{E}\left[|\delta Y_{t_k}|^2\right] + \mathrm{h}_n\int_{t_n}^{T}\mathbb{E}\left[|\delta Z_s|^2\right]ds\Big).\end{aligned}$$

Denoting $e_n := \mathbb{E}\left[|\delta Y_{t_n}|^2\right]$, $E_n := \max_{n\le k\le N} e_k$, $f_n := \int_{t_n}^{t_{n+1}}\mathbb{E}\left[|\delta Z_s|^2\right]ds$, $F_n = \sum_{k=n}^{N-1} f_k = \int_{t_n}^{T}\mathbb{E}\left[|\delta Z_s|^2\right]ds$ and $\varepsilon_n = (1+C\mathrm{h}_n)(\mathrm{h}_n\mathrm{h} + \int_{t_n}^{t_{n+1}}\mathbb{E}\left[|Z_s - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2\right]ds)$, the above writes, for any $0 \le n < N$,

$$e_n + \frac{1}{2}f_n \le (1+C\mathrm{h}_n)(e_{n+1} + \mathrm{h}_n E_n + \mathrm{h}_n F_n) + \varepsilon_n,$$

with $\sum_{n=0}^{N-1}\varepsilon_n \le C\mathrm{h}$ using $\sum_{n=0}^{N-1}\mathrm{h}_n = T$ and Lemma 4.2.

(Convergence estimate.) The extended Gronwall lemma 4.6 then yields

$$\max_{0\le n\le N}\mathbb{E}\left[|\delta Y_{t_n}|^2\right] + \int_0^T\mathbb{E}\left[|\delta Z_s|^2\right]ds \le C\left(\mathbb{E}\left[|\delta Y_T|^2\right] + \mathrm{h}\right) \le C\mathrm{h}, \tag{45}$$

where we used Lemma 4.3.

To conclude, coming back to (43) and using (45) and Lemma 4.2, we obtain, for each $t \in [0,T]$,

$$\mathbb{E}\left[|\delta Y_t|^2\right] \le C\mathrm{h}.$$

Finally, using Lemma 4.3 to obtain $\sup_{t\in[T,T+\theta]} \mathbb{E}\left[|\delta Y_t|^2\right] + \int_T^{T+\theta} \mathbb{E}\left[|\delta Z_t|^2\right] dt \le C\mathrm{h}$ allows to conclude the proof. □

Remark 4.3 (Piecewise constant reconstruction). Under the hypotheses of Theorem 4.7, set $Y_t^{\mathrm{pc}} = Y_{t_n}^{\mathrm{h}}$ and $Z_t^{\mathrm{pc}} = Z_{t_n}^{\mathrm{h}}$ on $[t_n, t_{n+1})$, with $Y_T^{\mathrm{pc}} = Y_T^{\mathrm{h}}$. Then

$$\sup_{t\le T} \mathbb{E}|Y_t - Y_t^{\mathrm{pc}}|^2 + \mathbb{E}\int_0^T |Z_t - Z_t^{\mathrm{pc}}|^2 dt \le C\mathrm{h}.$$

For Y, write, for $t \in [t_n, t_{n+1})$,

$$\mathbb{E}|Y_t - Y_t^{\mathrm{pc}}|^2 \le 2\mathbb{E}|Y_t - Y_{t_n}|^2 + 2\mathbb{E}|Y_{t_n} - Y_{t_n}^{\mathrm{h}}|^2.$$

The second term is bounded by $C\mathrm{h}$ by Theorem 4.7, since $Y_{t_n}^{\mathrm{h}} = \overline{Y}_{t_n}^{\mathrm{h}}$. For the first term, $Y_t = u(t, X_t)$ and the regularity of u in Theorem 3.6 give

$$\mathbb{E}|Y_t - Y_{t_n}|^2 \le C\big(t - t_n + \mathbb{E}|X_t - X_{t_n}|^2\big) \le C\mathrm{h},$$

where $\mathbb{E}|X_t - X_{t_n}|^2 \le C(t - t_n)$ follows from Assumption 3.1(iv). For Z, let $\widetilde{Z}^{\mathrm{h}}$ be the conditional average defined in (33). By (39) and Jensen's inequality,

$$\sum_{n=0}^{N-1} \mathrm{h}_n \mathbb{E}|\widetilde{Z}_{t_n}^{\mathrm{h}} - Z_{t_n}^{\mathrm{h}}|^2 \le \mathbb{E}\int_0^T |Z_t - \overline{Z}_t^{\mathrm{h}}|^2 dt.$$

Thus, writing $Z_t - Z_t^{\mathrm{pc}} = (Z_t - \widetilde{Z}_{t_n}^{\mathrm{h}}) + (\widetilde{Z}_{t_n}^{\mathrm{h}} - Z_{t_n}^{\mathrm{h}})$, the bound follows from $|a+b|^2 \le 2|a|^2 + 2|b|^2$, Lemma 4.2, and Theorem 4.7.

5 Peng and Yang (2009)'s examples

We compute the scheme of Section 4.3 analytically for two one-dimensional examples, with $X = W$. Assume $\theta > 0$ and $\theta/T \in \mathbb{Q}$, and choose a uniform step $\mathrm{h} > 0$ such that $N = T/\mathrm{h}$ and $\ell = \theta/\mathrm{h}$ are integers. Write $t_n = n\mathrm{h}$, $\Delta W_n = W_{t_{n+1}} - W_{t_n}$, and $\varepsilon \sim \mathcal{N}(0,1)$.

5.1 First example: anticipation of Y

The first example is Example 4.3 of Peng and Yang (2009):

$$Y_t = TW_T - \int_t^T \frac{\mathbb{E}_s[Y_{s+\theta}]}{s+\theta}\, ds - \int_t^T Z_s\, dW_s, \qquad 0 \le t \le T,$$

with terminal functions $\Phi(t,x) = tx$ and $\Psi(t,x) = t$ on $[T, T+\theta]$. Its solution is $Y_t = tW_t$, $Z_t = t$. The generator f of (21) and the anticipated laws are

$$f(t,x,y,z,m) = -\frac{1}{t+\theta}\int_{\mathbb{R}} a\, m(da), \qquad m_t(Y,Z) = \mathcal{L}_t(Y_{t+\theta}),$$

and the map in Assumption 3.2 is

$$m^{u,v}(t,x) = \mathcal{L}\Big(u(t+\theta, x + \sqrt{\theta}\,\varepsilon)\Big).$$

The uniform time-regularity conditions in Assumption 4.1(i), (iii) fail: the time increments of Φ and f cannot be bounded independently of x and m, respectively.

PDE. Equation (30) becomes

$$\partial_t u(t,x) + \frac{1}{2}\partial_{xx} u(t,x) - \frac{\mathbb{E}[u(t+\theta, x+\sqrt{\theta}\,\varepsilon)]}{t+\theta} = 0, \qquad u = \Phi \text{ on } [T, T+\theta]. \tag{46}$$

For $u(t,x) = tx$, the first two terms sum to x and the last term is $-x$. Moreover, $d(tW_t) = W_t dt + t\, dW_t$ verifies the BSDE directly.

Scheme and error. With $m^{\mathrm{h}}_{t_n} = \mathcal{L}_{t_n}(Y^{\mathrm{h}}_{t_{n+\ell}})$, the scheme reads

$$Z^{\mathrm{h}}_{t_n} = \frac{1}{\mathrm{h}}\mathbb{E}_{t_n}[Y^{\mathrm{h}}_{t_{n+1}} \Delta W_n], \qquad Y^{\mathrm{h}}_{t_n} = \mathbb{E}_{t_n}[Y^{\mathrm{h}}_{t_{n+1}}] - \frac{\mathrm{h}}{t_n+\theta}\mathbb{E}_{t_n}[Y^{\mathrm{h}}_{t_{n+\ell}}].$$

Starting from the prescribed terminal values, backward induction gives

$$\mathbb{E}_{t_n}[Y^{\mathrm{h}}_{t_{n+1}}] = t_{n+1} W_{t_n}, \qquad \mathbb{E}_{t_n}[Y^{\mathrm{h}}_{t_{n+\ell}}] = (t_n+\theta) W_{t_n},$$

and hence $Y^{\mathrm{h}}_{t_n} = t_n W_{t_n}$ and $Z^{\mathrm{h}}_{t_n} = t_{n+1}$ for $n < N$. The martingale representation of $t_{n+1}W_{t_{n+1}}$ gives $\overline{Z}^{\mathrm{h}}_t = t_{n+1}$ on $[t_n, t_{n+1})$. For $t = t_n + s$, $0 \le s < \mathrm{h}$, reconstruction (38) therefore yields

$$Z_t - \overline{Z}^{\mathrm{h}}_t = s - \mathrm{h}, \qquad Y_t - \overline{Y}^{\mathrm{h}}_t = (s-\mathrm{h})(W_t - W_{t_n}).$$

Since $\mathbb{E}|W_t - W_{t_n}|^2 = s$, the errors are

$$\mathbb{E}\int_0^T |Z_t - \overline{Z}^{\mathrm{h}}_t|^2 dt = \frac{T\mathrm{h}^2}{3}, \qquad \sup_{t\le T}\mathbb{E}|Y_t - \overline{Y}^{\mathrm{h}}_t|^2 = \frac{4\mathrm{h}^3}{27}.$$

5.2 Second example: anticipation of Z

The second example is taken from Example 5.3 of Peng and Yang (2009):

$$Y_t = W_T^2 - T - \int_t^T \sqrt{\frac{\pi}{2\theta}}\,\mathbb{E}_s[|Z_{s+\theta} - Z_s|]\, ds - \int_t^T Z_s\, dW_s, \qquad 0 \le t \le T,$$

with terminal functions $\Phi(t,x) = x^2 + t - 2T$ and $\Psi(t,x) = 2x$ on $[T, T+\theta]$. Its solution is $Y_t = W_t^2 + t - 2T$, $Z_t = 2W_t$. The generator f of (21) and the anticipated laws are

$$f(t,x,y,z,m) = -\sqrt{\frac{\pi}{2\theta}}\int_{\mathbb{R}} |a - z|\, m(da), \qquad m_t(Y,Z) = \mathcal{L}_t(Z_{t+\theta}),$$

and the map in Assumption 3.2 is

$$m^{u,v}(t,x) = \mathcal{L}\Big(v(t+\theta, x+\sqrt{\theta}\,\varepsilon)\Big).$$

The terminal function is not globally Lipschitz in x, so Assumption 3.1(i) fails.

PDE. Equation (30) becomes

$$\begin{aligned}&\partial_t u(t,x) + \frac{1}{2}\partial_{xx} u(t,x)\\ &\quad - \sqrt{\frac{\pi}{2\theta}}\,\mathbb{E}\Big[\Big|\partial_x u(t+\theta, x+\sqrt{\theta}\,\varepsilon) - \partial_x u(t,x)\Big|\Big] = 0,\end{aligned} \tag{47}$$

with $u = \Phi$ on $[T, T+\theta]$. For $u(t,x) = x^2 + t - 2T$, the differential terms sum to 2, while

$$\sqrt{\frac{\pi}{2\theta}}\,\mathbb{E}|2\sqrt{\theta}\,\varepsilon| = 2.$$

Also, $d(W_t^2 + t - 2T) = 2dt + 2W_t\, dW_t$ verifies the BSDE.

Scheme and error. With $m^{\mathrm{h}}_{t_n} = \mathcal{L}_{t_n}(Z^{\mathrm{h}}_{t_{n+\ell}})$, the scheme reads

$$Z^{\mathrm{h}}_{t_n} = \frac{1}{\mathrm{h}}\mathbb{E}_{t_n}[Y^{\mathrm{h}}_{t_{n+1}}\Delta W_n], \qquad Y^{\mathrm{h}}_{t_n} = \mathbb{E}_{t_n}[Y^{\mathrm{h}}_{t_{n+1}}] - \mathrm{h}\sqrt{\frac{\pi}{2\theta}}\,\mathbb{E}_{t_n}|Z^{\mathrm{h}}_{t_{n+\ell}} - Z^{\mathrm{h}}_{t_n}|.$$

Starting from the terminal values, backward induction gives

$$Z^{\mathrm{h}}_{t_n} = 2W_{t_n}, \qquad \mathbb{E}_{t_n}[Y^{\mathrm{h}}_{t_{n+1}}] = Y_{t_n} + 2\mathrm{h}, \qquad \mathbb{E}_{t_n}|Z^{\mathrm{h}}_{t_{n+\ell}} - Z^{\mathrm{h}}_{t_n}| = 2\sqrt{\frac{2\theta}{\pi}}.$$

Thus $Y^{\mathrm{h}}_{t_n} = Y_{t_n}$ at every grid date. Finally,

$$Y_{t_{n+1}} - \mathbb{E}_{t_n}[Y_{t_{n+1}}] = W^2_{t_{n+1}} - W^2_{t_n} - \mathrm{h} = \int_{t_n}^{t_{n+1}} 2W_s\,dW_s.$$

Consequently, reconstruction (38) satisfies $\overline{Z}^{\mathrm{h}} = Z$ and $\overline{Y}^{\mathrm{h}} = Y$: both errors vanish.

6 Agarwal et al. (2019)'s MVA ABSDE

Assuming Black–Scholes dynamics

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

the risk-neutral price process V^{bs} of an option with payoff $\Phi(S_T)$ solves (assuming constant drifts μ above and risk-free rate r below)

$$V^{bs}_t = \Phi(S_T) + \int_t^T \left[Z^{bs}_s\sigma^{-1}(r-\mu) - rV^{bs}_s\right] ds - \int_t^T Z^{bs}_s dW_s, \tag{48}$$

for some progressive integrable process Z^{bs}, while the value process accounting for Agarwal et al. (2019)'s MVA (of an ES initial margin) solves

$$V_t = \Phi(S_T) + \int_t^T \left[Z_s\sigma^{-1}(r-\mu) + R\,\mathbb{ES}^{\alpha}_s\big(V_s - V_{(s+\Delta)\wedge T}\big) - rV_s\right] ds - \int_t^T Z_s dW_s, \tag{49}$$

for some progressive integrable process Z. Hence their $\mathrm{MVA}_t = V_t - V^{bs}_t$ solves

$$\begin{aligned}\mathrm{MVA}_t = \int_t^T \Big[&\zeta_s\sigma^{-1}(r-\mu) + R\,\mathbb{ES}^{\alpha}_s\big(V^{bs}_s - V^{bs}_{(s+\Delta)\wedge T} + \mathrm{MVA}_s - \mathrm{MVA}_{(s+\Delta)\wedge T}\big)\\ &- r\mathrm{MVA}_s\Big] ds - \int_t^T \zeta_s dW_s,\end{aligned} \tag{50}$$

for some progressive integrable process $\zeta(= Z - Z^{bs})$. In this example, the unknowns (Y, Z) of our scheme are therefore (MVA, ζ), with terminal condition $\mathrm{MVA}_T = 0$. The Black–Scholes price V^{bs} is known analytically. Writing $\bar{t} = (t+\Delta)\wedge T$ and $m_t = \mathcal{L}(-V_{\bar{t}} \mid \mathcal{F}_t)$, conditional cash invariance identifies the generator of (50) as

$$f(t,x,y,z,m) = (R-r)y + \frac{r-\mu}{\sigma}z + R\left[v^{bs}(t,x) + \mathbb{ES}^{\alpha}(m)\right]. \tag{51}$$

Here $v^{bs}(t,x)$ is the analytic Black–Scholes price at time t and spot x, y and z represent the MVA and its control ζ, and $\mathbb{ES}^{\alpha}(m)$ is the expected shortfall of a loss with law m. The dependence of the conditional risk measure on the future value of the solution makes (50) (like (49) in the first place) an anticipated, nonlinear equation. We use $S_0 = K = 20$, $T = 1$, $r = \mu = R = 0.02$, $\sigma = 0.25$, $\alpha = 0.99$, and $\Delta = 0.02$, as in their numerical study. The margin period Δ is fixed

within each experiment and is distinct from the numerical time step h; all our grids are chosen so that Δ/h is an integer.

The Markov property yields

$$\begin{aligned} V_t^{bs} &= v^{bs}(t,S_t), & Z_t^{bs} &= \sigma S_t \partial_S v^{bs}(t,S_t),\\ V_t &= v(t,S_t), & Z_t &= \sigma S_t \partial_S v(t,S_t),\\ \mathrm{MVA}_t &= \mathrm{mva}(t,S_t), & \zeta_t &= \sigma S_t \partial_S \mathrm{mva}(t,S_t). \end{aligned}$$

For $t = t_n = n\mathrm{h}$, let

$$\bar{n} = \min\{n + \Delta/\mathrm{h}, N\}, \qquad \bar{t} = t_{\bar{n}} = (t+\Delta) \wedge T, \qquad \Delta_t = \bar{t} - t.$$

Since Δ/h is an integer, no projection onto the time grid is needed. Let

$$c_t(x) = \mathbb{ES}^\alpha\big(-V_{\bar{t}} \mid S_t = x\big).$$

By conditional cash invariance of expected shortfall (Artzner et al., 1999),

$$\mathbb{ES}_t^\alpha(V_t - V_{\bar{t}}) = V_t + c_t(S_t).$$

At any grid time $t = n\mathrm{h}$, we denote by $\widehat{\mathrm{mva}}_n$ the Hermite approximation of the MVA function $\mathrm{mva}(t,\cdot)$. For a spot value x, define

$$\widehat{v}_n(x) := v^{bs}(t,x) + \widehat{\mathrm{mva}}_n(x),$$

and then let $\widehat{V}_n := \widehat{v}_n(S_t)$. At maturity, $\widehat{\mathrm{mva}}_N = 0$ and $\widehat{v}_N = \Phi$. When constructing the conditional-ES target at time t, we use this already computed approximation function at the future date $\bar{t}$. Conditionally on $S_t = x$, write the Black–Scholes transition as

$$S_{t_{\bar{n}}} = x \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)\Delta_t + \sigma\sqrt{\Delta_t}\,\varepsilon\right),$$

where $\varepsilon \sim \mathcal{N}(0,1)$. Let $S_{\bar{t}}^{t,x}$ denote this transition started from x at time t. For any candidate MVA function u giving a square-integrable loss, the map in Assumption 3.2 is

$$m^u(t,x) = \mathcal{L}\Big(-v^{bs}(\bar{t}, S_{\bar{t}}^{t,x}) - u(\bar{t}, S_{\bar{t}}^{t,x})\Big).$$

It depends only on (t,x) and u, since the anticipated law does not involve the control ζ.

Associated PDE Substituting the generator and $\zeta = \sigma x \partial_x \mathrm{mva}$ into (30) yields, for $t < T$ and $x > 0$,

$$\begin{aligned} 0 = \partial_t \mathrm{mva} + \frac{1}{2}\sigma^2 x^2 \partial_{xx}\mathrm{mva} + rx\partial_x \mathrm{mva}\\ + (R-r)\mathrm{mva} + R[v^{bs}(t,x) + c_t(x)], \end{aligned} \tag{52}$$

with $\mathrm{mva}(T,x) = 0$. Here $c_t(x) = \mathbb{ES}^\alpha(-v(\bar{t}, S_{\bar{t}}^{t,x}))$ depends on the total future value $v = v^{bs} + \mathrm{mva}$, so this is a nonlocal equation, common to the call and butterfly. The local drift reduces to r, while the transition inside c_t retains drift μ. This identifies the associated PDE; its classical regularity and the time-regularity conditions of Assumption 4.1 are not established here.

Numerical approximation Let $F_{\mathcal{N}}$ be the standard-normal distribution function and $U \sim \mathcal{U}(0,1)$. For $u \in (0,1)$, set

$$\ell^{t,x}(u) = -\widehat{v}_{\overline{n}}\left(x \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)\Delta_t + \sigma\sqrt{\Delta_t}\, F_{\mathcal{N}}^{-1}(u)\right]\right).$$

For every bounded continuous test function ψ,

$$\mathbb{E}\Big[\psi(-\widehat{V}_{\overline{n}}) \mid S_t = x\Big] = \int_0^1 \psi(\ell^{t,x}(u))du \simeq \sum_{q=1}^{Q} p_q \psi(\ell^{t,x}(u_q))$$

where $(u_q, p_q)_{q=1}^{Q}$ are the Gauss–Legendre nodes and positive normalized weights on $(0,1)$. The equality is exact for the fitted future value because the conditional Black–Scholes transition is represented exactly by $U \mapsto F_{\mathcal{N}}^{-1}(U)$. Replacing $V_{\overline{t}}$ by $\widehat{V}_{\overline{n}}$ includes the time-discretization and regression errors; the remaining approximation in the display is solely the Gauss–Legendre quadrature error. The conditional loss law is then approximated by the discrete probability measure

$$\mu_{t,x}^{Q} = \sum_{q=1}^{Q} p_q \delta_{\ell^{t,x}(u_q)}.$$

Using the variational representation of Rockafellar and Uryasev (2000),

$$c_t(x) = \min_{a\in\mathbb{R}}\left\{a + \frac{1}{1-\alpha}\mathbb{E}\Big[\big(-V_{\overline{t}} - a\big)^+ \mid S_t = x\Big]\right\}.$$

After replacing $V_{\overline{t}}$ by $\widehat{V}_{\overline{n}}$ and applying quadrature, we obtain

$$\widehat{c}_n^{Q}(x) = \min_{a\in\mathbb{R}}\left\{a + \frac{1}{1-\alpha}\sum_{q=1}^{Q} p_q\big(\ell^{t,x}(u_q) - a\big)^+\right\}.$$

For both payoffs, we use the complete measure $\mu_{t,x}^{Q}$, sort its losses in decreasing order, and average them with their weights until the accumulated mass reaches $1-\alpha$, using only the required fraction of the final weight. We approximate the function $x \mapsto v^{bs}(t_n, x) + c_{t_n}(x)$ in the generator (51) by Hermite regression of the targets $v^{bs}(t_n, S_{t_n}^{(m)}) + \widehat{c}_n^{Q}(S_{t_n}^{(m)})$.

At each time step, the Hermite least-squares procedure of Appendix A estimates the following conditional expectations:

$$\widetilde{\mathrm{MVA}}_n = \mathbb{E}_n[\mathrm{MVA}_{n+1}], \qquad \zeta_n = \frac{1}{h}\mathbb{E}_n\Big[(\mathrm{MVA}_{n+1} - \widetilde{\mathrm{MVA}}_n)\Delta W_n\Big].$$

The backward scheme for (50) combines these estimates to form the regression targets for the MVA at t_n. A final Hermite regression provides its approximation for use at earlier dates. The recursion starts from $\mathrm{MVA}_N = 0$. We reconstruct $V_n = V_n^{bs} + \mathrm{MVA}_n$ and $Z_n = Z_n^{bs} + \zeta_n$ whenever required. In particular, the ES is computed from the total future value.

Proxies For comparison, Agarwal et al. (2019) replace the short-horizon value variation by its frozen-delta approximation

$$V_s - V_{(s+\Delta)\wedge T} \simeq -|Z_s|\sqrt{(s+\Delta)\wedge T - s}\, G, \qquad G \sim \mathcal{N}(0,1),$$

which gives the standard nonlinear BSDE

$$V_t^{nl} = \Phi(S_T) + \int_t^T \Big[-rV_s^{nl} + Z_s^{nl}\sigma^{-1}(r-\mu) + RC_\alpha\sqrt{(s+\Delta)\wedge T - s}\,|Z_s^{nl}|\Big]\, ds - \int_t^T Z_s^{nl} dW_s.$$

where $C_\alpha = \mathbb{ES}^\alpha(G)$. Their linear approximation replaces Z^{nl} in the source term by the Black–Scholes hedge Z^{bs}; setting $R = 0$ gives the usual no-initial-margin Black–Scholes price. Thus the nonlinear and linear equations are small-Δ proxies, whereas our first computation targets (50) directly.

The proxy PDEs replace the full risk $v(t,x) + c_t(x)$ respectively by

$$C_\alpha \sqrt{\Delta_t}\, \sigma x |\partial_x v^{nl}(t,x)| \quad \text{and} \quad C_\alpha \sqrt{\Delta_t}\, \sigma x |\partial_x v^{bs}(t,x)|,$$

where v^{nl} is the nonlinear proxy price function.

Both proxy equations have lost the anticipated character of the original MVABSDE: their drivers only depend on quantities available at the current date. Accordingly, our time discretization reduces naturally in these cases to the implicit scheme of Bouchard and Touzi (2004). We use the same Hermite regression procedure, without the conditional-ES step. The nonlinear proxy evaluates its driver with the reconstructed control $|Z_n^{bs} + \zeta_n|$, whereas the linear proxy uses the analytic $|Z_n^{bs}|$. Neither proxy requires an ES regression or a Gauss–Legendre quadrature; these are needed only for the genuine MVABSDE scheme.

Numerical results Tables 1 and 2 report the MVA percentage, defined as

$$100 \frac{\mathrm{MVA}_0}{V_0^{bs}} = 100 \frac{V_0 - V_0^{bs}}{V_0^{bs}} \quad (\%).$$

This expresses the price adjustment as a percentage of the Black–Scholes price. They do not report the total price V_0, nor the ES collateral itself. Each Monte Carlo entry is the mean over ten independent runs, followed by its sample standard deviation. This dispersion measures the combined Monte Carlo and regression variability, not the regression error alone. We use 200 000 common training paths per run and Hermite degree two. Both payoffs use $Q = 256$ full-law quadrature nodes.

The "deterministic reference" column is a same-grid validation of the direct solver. It uses the same time grid, the same Black–Scholes control-variate decomposition, the same implicit backward update, and the same Gauss–Legendre construction of the conditional ES. The only change is how the conditional functions are evaluated. The Monte Carlo solver regresses them on the Hermite basis from simulated paths; the deterministic solver stores them on a fine log-spot grid and integrates the one-step Gaussian transition by Gauss–Hermite quadrature, using interpolation between grid points.

The "Article" row converts the proxy prices reported in Agarwal et al. (2019, Remark 4.1 and Tables 1, 3, and 5) to the same MVA percentage. These are proxy-specific reference values; the cited article does not provide a direct-MVABSDE value.

Table 1: ATM call: MVA percentage.

N	direct MVABSDE	deterministic reference	nonlinear proxy	linear proxy
50	$0.861560 \pm 1.31 \times 10^{-3}$	0.861569	$1.010607 \pm 1.32 \times 10^{-3}$	$1.007173 \pm 1.32 \times 10^{-3}$
100	$0.859312 \pm 9.07 \times 10^{-4}$	0.859493	$1.007756 \pm 9.32 \times 10^{-4}$	$1.004309 \pm 9.30 \times 10^{-4}$
200	$0.858423 \pm 1.10 \times 10^{-3}$	0.858241	$1.006538 \pm 1.14 \times 10^{-3}$	$1.003083 \pm 1.14 \times 10^{-3}$
Article	–	–	1.002170	1.057365

At $N = 200$ (Tables 1 and 2), the MVA of the original MVABSDE is 14.7% and 14.4% below the nonlinear and linear call proxies, respectively. For the butterfly, it is 5.5% and 5.7% above the corresponding proxy MVAs. By comparison, the Hermite MVA differs from the deterministic reference by only 0.021% for the call and 0.045% for the butterfly, relative to the reference MVA.

Robustness with respect to the conditional-risk estimator. At $N = 200$, we compare three ways of approximating the conditional ES used to construct the MVA regression targets:

Table 2: ATM butterfly: MVA percentage.

N	direct MVABSDE	deterministic reference	nonlinear proxy	linear proxy
50	$0.765830 \pm 7.52 \times 10^{-4}$	0.766698	$0.724722 \pm 6.56 \times 10^{-4}$	$0.723675 \pm 6.56 \times 10^{-4}$
100	$0.773317 \pm 6.21 \times 10^{-4}$	0.773731	$0.732213 \pm 5.26 \times 10^{-4}$	$0.731134 \pm 5.27 \times 10^{-4}$
200	$0.775954 \pm 4.33 \times 10^{-4}$	0.776302	$0.735188 \pm 2.07 \times 10^{-4}$	$0.734097 \pm 2.07 \times 10^{-4}$
Article	–	–	0.700663	0.700663

Gauss–Legendre quadrature with the Rockafellar–Uryasev formula, nested Monte Carlo with the Rockafellar–Uryasev formula, and the two-stage VaR/ES learning procedure of Barrera et al. (2026). All three use the same time-discrete MVABSDE scheme, simulated paths, and degree-two Hermite regressors for the MVA and ζ. Only the conditional-ES estimator changes. Table 3 reports the resulting MVA percentages.

Table 3: Direct MVABSDE at $N = 200$: MVA percentage, reported as mean ± sample standard deviation over three common-path runs.

Conditional-risk estimator	ATM call	ATM butterfly
Gauss–Legendre + Rockafellar–Uryasev	$0.858587 \pm 1.49 \times 10^{-3}$	$0.775644 \pm 2.61 \times 10^{-4}$
Nested Monte Carlo	$0.859423 \pm 1.21 \times 10^{-3}$	$0.777843 \pm 7.81 \times 10^{-3}$
Conditional VaR/expected-shortfall learning	$0.860684 \pm 2.61 \times 10^{-3}$	$0.785767 \pm 4.15 \times 10^{-3}$

Relative to the quadrature MVA, the largest differences between mean MVAs are 0.24% for the call and 1.31% for the butterfly. These are smaller than the discrepancies with the proxies, supporting their robustness at this time grid, without excluding a shared time-discretization bias.

Larger margin periods. To test the small-Δ approximation of Agarwal et al. (2019, Theorem 3.1), we increase Δ while keeping the other financial parameters fixed. Table 4 uses $N = 200$, 200 000 paths per run, degree-two Hermite regression and the same quadrature rules as above. These larger horizons are stress tests of the proxies, not standard margin periods. All sample standard deviations are below 0.006 percentage points.

Table 4: MVA percentage: means over three independent runs, with Δ in years.

Product	Δ	Direct MVABSDE	Nonlinear proxy	Linear proxy
Call	0.02	0.8586	1.0066	1.0032
	0.10	1.5177	2.2000	2.1836
	0.50	1.8903	4.2725	4.2114
Butterfly	0.02	0.7756	0.7352	0.7341
	0.10	1.4921	1.5059	1.5016
	0.50	2.0144	2.4681	2.4592

For the call, the direct MVA is about 15% below the nonlinear proxy at $\Delta = 0.02$, versus 56% at $\Delta = 0.50$. For the butterfly, the gap changes sign: from about 5.5% above to 18.4% below, so its magnitude need not increase monotonically. Checks at $N = 400$ and refinement of the deterministic reference in space and quadrature preserve the large-Δ gaps. The direct scheme avoids the small-Δ approximation, but these checks do not establish a uniform-in-Δ numerical error bound.

Unlike the explicit examples of Section 5, no exact direct solution is available here, and a time-zero price alone does not give the full error in (MVA, ζ). For the call, the change between the two finest grids remains of the same order as the across-seed variability. For the butterfly this variability is smaller, but the three time grids still do not provide a stable asymptotic slope

against an exact target. The experiment therefore assesses numerical feasibility and fixed-grid accuracy rather than providing a direct numerical test of the $\mathrm{O}(h)$ error estimate in Theorem 4.7. An $\mathrm{O}(h)$ rate is likewise not identified in the numerical study of Abbas-Turki et al. (2025): they report empirical stability under time refinement and leave the time-consistency analysis of their regression-based scheme for future work.

7 Conclusion

In this paper, we introduce a new class of anticipated backward stochastic differential equations (ABSDEs) in a Brownian filtration, and we have proved existence and uniqueness of solutions under Lipschitz assumptions on both the driver and the anticipated operator, in a non-Markovian framework. In a Markovian framework, we obtain the representation of the solution (Y, Z) as deterministic functions of the time and of the Markovian state. We introduce an approximation scheme for scalar Markovian ABSDEs and proved its convergence under suitable regularity and consistency assumptions.We also prove that any sufficiently regular solution of the associated nonlocal PDE represents the Markovian ABSDE solution.

The analytical study of the PDEs associated with Markovian ABSDEs, in particular the important question of comparison and uniqueness of viscosity solutions, as well as their regularity under suitable assumptions on the coefficients, is left for further research. Even in the classical semilinear setting, comparison proofs for viscosity solutions typically rely on the doubling-of-variables technique and Ishii's lemma (see e.g. Crandall et al. (1992)). In the present setting, the PDE (30) is nonlocal in time: the equation at (t, x) involves the values of the solution on the entire future domain $[t, T + \theta] \times \mathbb{R}^d$, creating an obstruction to the classical doubling-of-variables technique used to prove comparison principles for viscosity solutions. A possible strategy to overcome this difficulty would be to work backward in time on intervals of size at most δ, assuming the anticipated term at any time t introduces a dependence only upon the future interval $[t, t + \theta]$, and that the anticipated term induces only a small perturbation from a classical non-anticipated driver. This analysis is beyond the scope of the present paper and will be addressed in future work.

This work suggests several other directions for future research. First, it would be interesting to relax the Lipschitz assumptions on the coefficients, and in particular to consider quadratic growth in Z. Second, it would be interesting to extend our results to the case of ABSDEs also driven by a jump process. In our numerical experiments, space discretization is handled by local least-squares regressions on a fixed Hermite basis. The resulting approximation error is distinct from the time-discretization error analysed in this paper. The additional impact of this regression-error could be analysed as well. Finally, it would be interesting to apply our results to other problems in economics and finance (beyond the XVA setup of Abbas-Turki et al. (2025)), such as optimal control problems with delay.

A Hermite regression

Let H_i denote the ith normalized probabilists' Hermite polynomial. For $t_n > 0$, set $U_n = W_{t_n}/\sqrt{t_n}$ and

$$P_\beta(u) = \sum_{i=0}^{d} \beta_i H_i(u), \qquad \beta \in \mathbb{R}^{d+1}.$$

Given a square-integrable target R_n, we approximate $\mathbb{E}[R_n \mid U_n]$ by $P_{\widehat{\beta}_n[R]}(U_n)$, where

$$\widehat{\beta}_n[R] = \arg\min_{\beta \in \mathbb{R}^{d+1}} \left\{ \sum_{m=1}^{M} \left| P_\beta(U_n^{(m)}) - R_n^{(m)} \right|^2 + \lambda |\beta|^2 \right\}, \qquad \lambda = 10^{-10}.$$

The normal equations are solved in double precision; the ridge penalty stabilizes the linear solve. At $t_0 = 0$, the fitted quantity is constant. Section 6 specifies the targets and uses degree $d = 2$.

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